

Assignment 2

Glasgow School data, Wave 1

2026-05-05

1. Geometrically Weighted Terms in ERGMs

The geometrically weighted out-degree (GWOD) statistic is

$$s_{GWD}(x, \alpha) = e^\alpha \sum_{k=1}^n \{1 - (1 - e^{-\alpha})^k\} D_k^{out}(x),$$

where $D_k^{out}(x)$ is the number of nodes with out-degree exactly k . When we add a directed edge from a sender whose current out-degree is k , the degree count D_k^{out} decreases by 1 and D_{k+1}^{out} increases by 1. All other degree counts remain unchanged. Therefore, the change statistic is:

$$\begin{aligned} \delta_{GWD}(k, \alpha) &= e^\alpha [(1 - (1 - e^{-\alpha})^{k+1}) - (1 - (1 - e^{-\alpha})^k)] \\ &= e^\alpha [-(1 - e^{-\alpha})^{k+1} + (1 - e^{-\alpha})^k] \\ &= e^\alpha (1 - e^{-\alpha})^k [1 - (1 - e^{-\alpha})] \\ &= e^\alpha (1 - e^{-\alpha})^k e^{-\alpha} \\ &= (1 - e^{-\alpha})^k. \end{aligned}$$

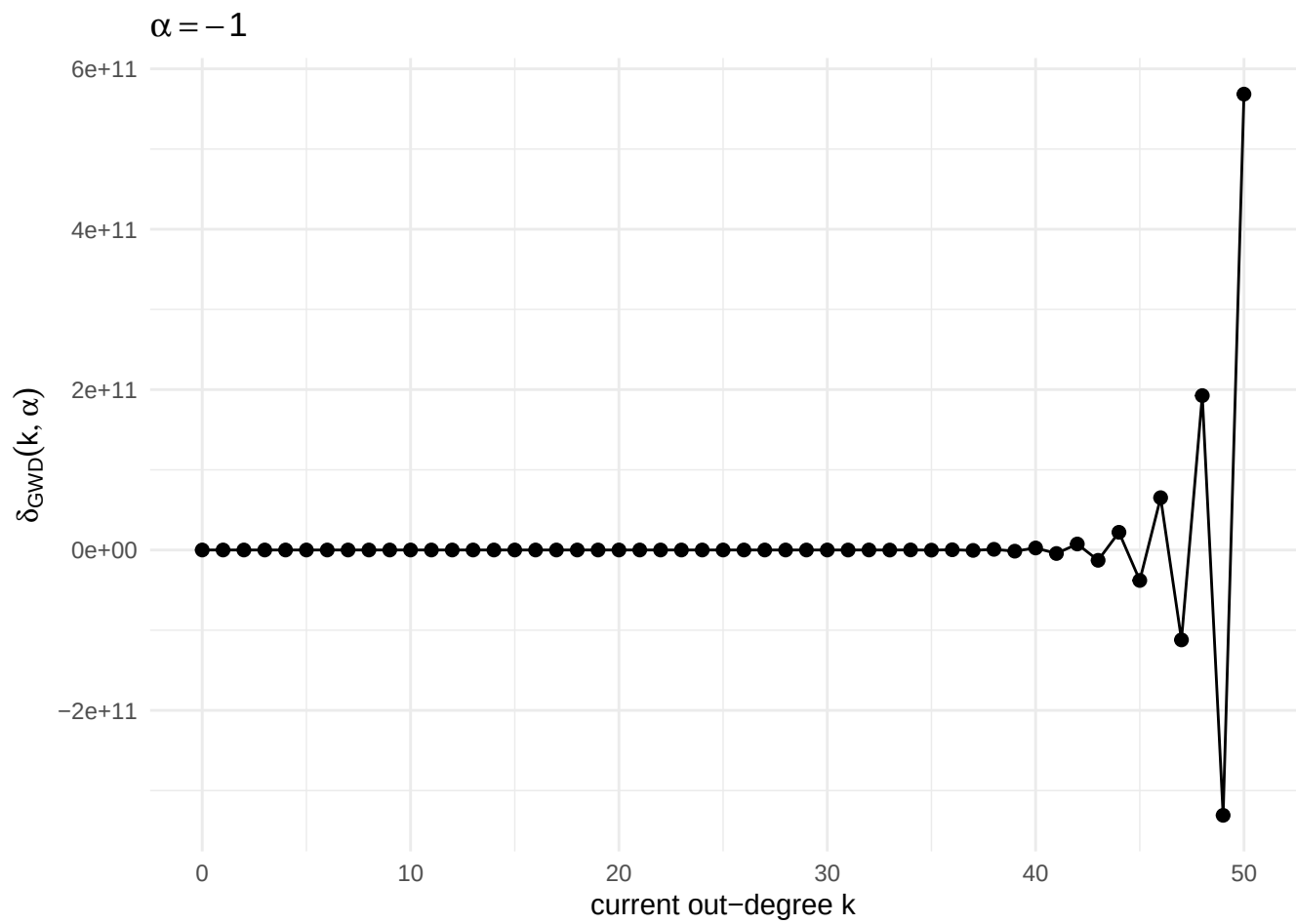
Thus, the change statistic is

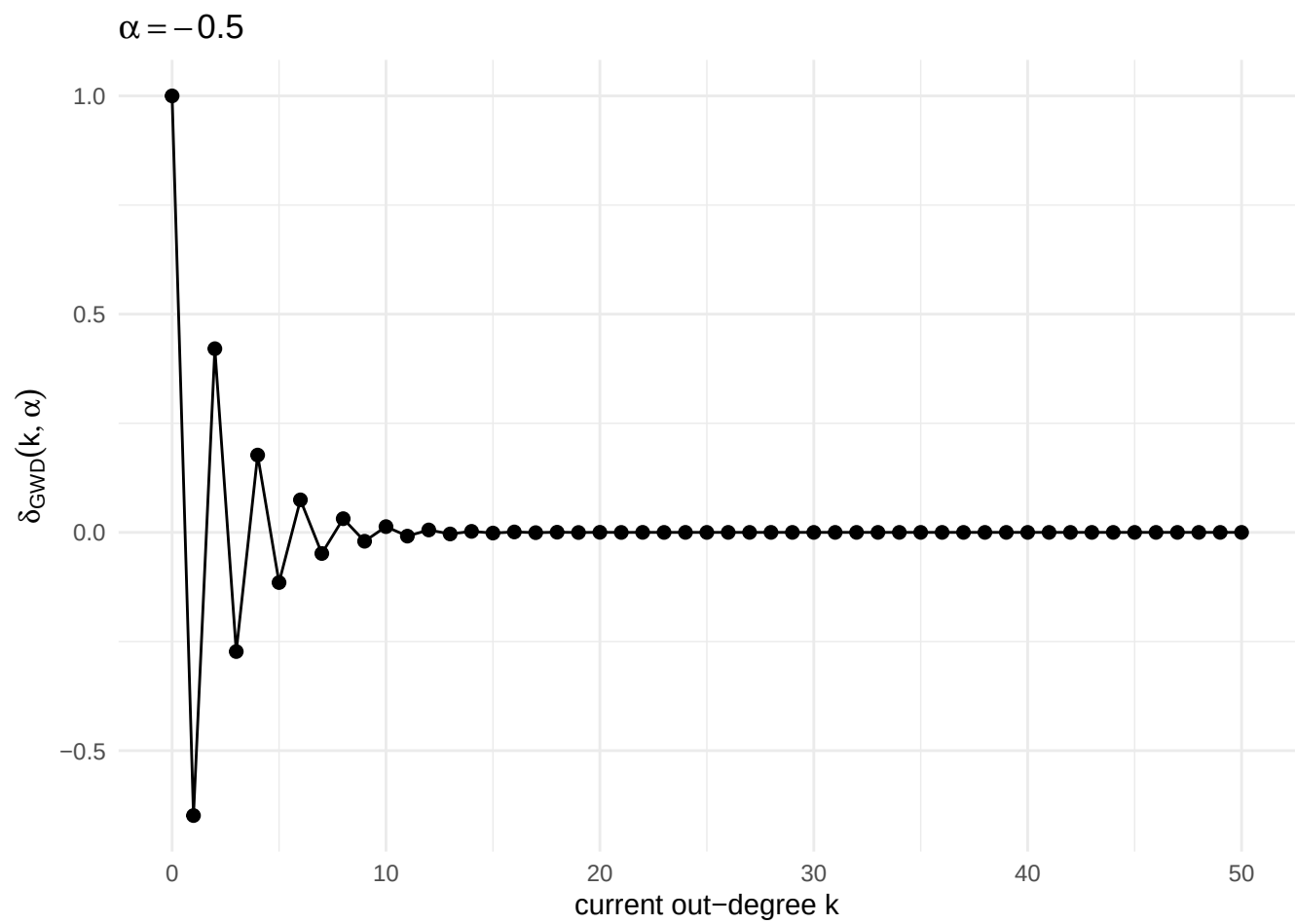
$$\delta_{GWD}(k, \alpha) = (1 - e^{-\alpha})^k.$$

```
# setup
delta_GWD <- function(k, alpha) {
  (1 - exp(-alpha))^k
}

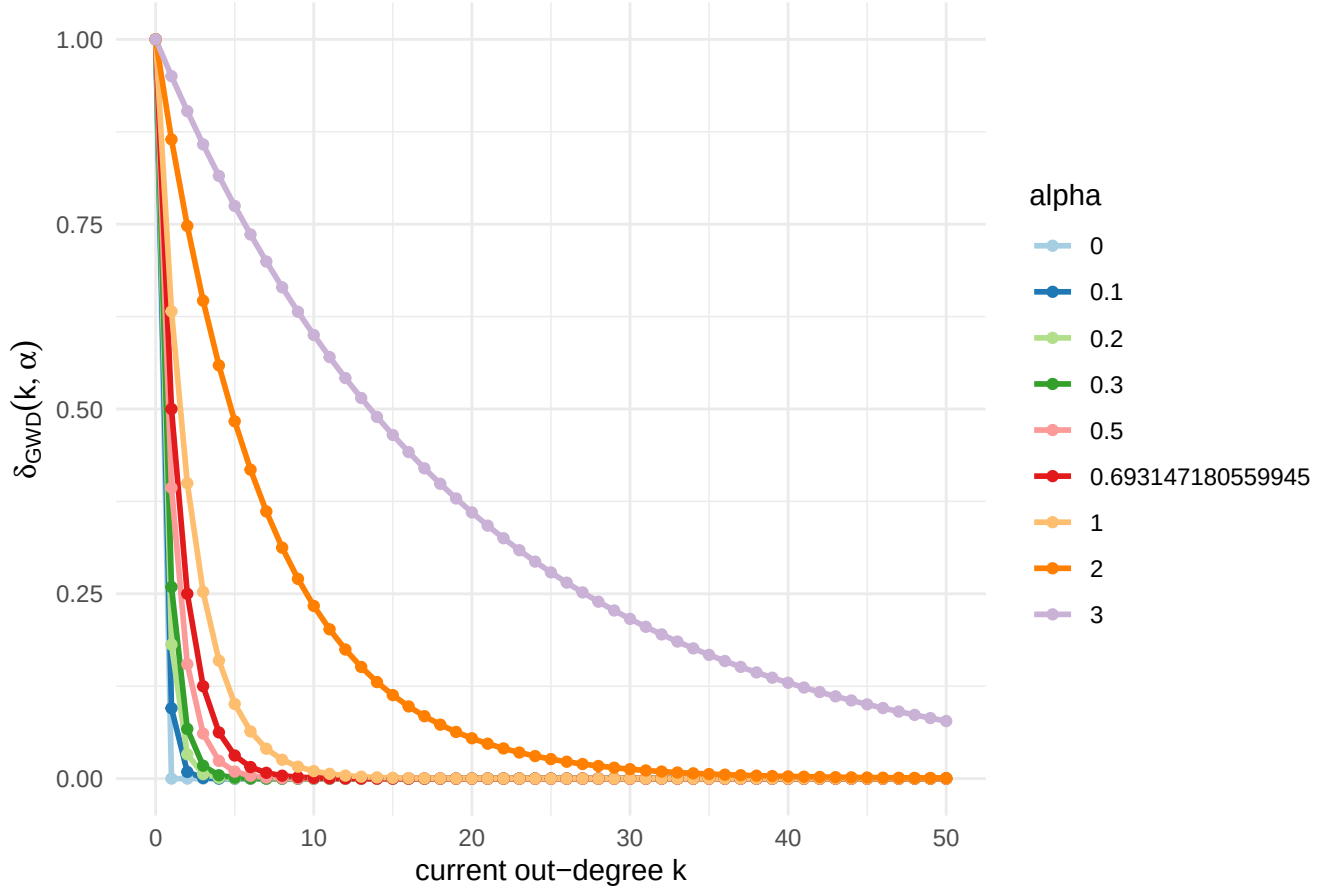
alpha_vals <- c(-1, -0.5, 0, 0.1, 0.2, 0.3, 0.5, log(2), 1, 2, 3)
neg_alpha <- c(-1, -0.5)
nonneg_alpha <- setdiff(alpha_vals, neg_alpha)

k_range <- 0:max_k
```





Change statistic for non.negative decay values



The change statistic

$$\delta(k, \alpha) = (1 - e^{-\alpha})^k$$

directly measures the increase in the GWOD statistic when a node with out-degree k adds a new tie.

Its behaviour depends critically on α :

For $\alpha > 0$, the base $b = 1 - e^{-\alpha}$ lies strictly between 0 and 1. Hence δ is a decreasing geometric sequence in k . * Very small positive α (e.g., 0.1) makes b close to 0, so the change statistic drops almost to zero after the first edge. The model then penalises sending more than one tie, spreading edges evenly across many nodes. * Medium values like $\ln(2)$ give $b = 0.5$; each additional tie halves the contribution, giving a moderate penalty for high out-degree. * Large α (2 or 3) pushes b close to 1, making $\delta \approx 1$ for all k . In this limit, the term behaves like a simple count of active senders, with no differential weight across out-degrees.

For $\alpha = 0$, then $b = 0$, so $\delta(0) = 1$ and $\delta(k) = 0$ for all $k \geq 1$. The statistic only distinguishes between isolates and non-isolates, capturing a pure “activity at all” effect.

For $\alpha < 0$, $e^{-\alpha} > 1$ so $b < 0$ and $|b| > 1$. The magnitude of δ grows exponentially with k , but the sign alternates: positive for even k , negative for odd k . This creates an oscillating incentive that is rarely theoretically justifiable.

Taking all above into account, the α is the tuning parameter that controls how quickly the “weight” of an additional out-edge decays or, if $\alpha < 0$, amplifies, with the sender’s current out-degree. Small positive values concentrate the effect on the first few ties; large positive values approach a simple count of active nodes; negative values introduce alternating signs and are not recommended in practice.

Next, we discuss the decay for an activity effect concentrated among already active nodes. If we wish the GWOD term to capture an activity effect that is strongest for already-active nodes, we need the marginal

contribution $\delta(k)$ to *increase* with k .

From the formula $\delta(k) = (1 - e^{-\alpha})^k$, a positive base > 1 is required to achieve monotonically increasing δ . However, $1 - e^{-\alpha} > 1$ would demand $e^{-\alpha} < 0$, which is impossible for real α .

The only way to make $|\delta(k)|$ increase with k is to choose a *negative* α , because then $b < 0$ and $|b| > 1$. For example, $\alpha = -1$ yields $\delta(k) \approx (-1.718)^k$, whose magnitude grows rapidly with k .

Superficially, this emphasises high out-degree actors. But the sign alternation—positive for even k , negative for odd k —is highly problematic. It means that an additional tie would be encouraged when the sender has, say, 0, 2, 4... ties and discouraged when it has 1, 3, 5... ties. Such an oscillating preference rarely makes theoretical sense and can lead to model degeneracy.

Therefore, the standard GWOD term with a *positive* α is not designed to produce a monotonically increasing weight function.

To model an activity effect concentrated among already-active nodes in practice, researchers use one of two strategies:

- Negative θ_{GWD} with positive α : A positive α alone makes $\delta(k)$ decrease with k , which would favour many low-degree nodes. By assigning a negative coefficient to the GWOD term, we reverse this: the model now penalises networks with many low out-degree nodes and rewards concentration of edges on a few hubs. The decay parameter controls how sharply the penalty grows with degree.
- Explicit sender-activity covariates: Terms like `nodecov` (main effect of a continuous attribute on out-degree) or `nodefactor` allow the model to directly capture that students with higher alcohol consumption or other activity-related traits send more ties.

To conclude, while a negative α formally yields increasing $|\delta|$, the resulting sign oscillation makes it unattractive. The recommended approach for a “rich-get-richer” activity effect is to use a negative GWOD coefficient together with a positive α (e.g., 0.3 – 0.7) or to include dedicated covariate effects.

Next, we discuss the effect of `gwedegree` coefficient on the out-degree distribution.

```
set.seed(1234)
m1 <- ergm(net ~ edges, control = control.ergm(seed = 123))
summary(m1)

## Call:
## ergm(formula = net ~ edges, control = control.ergm(seed = 123))
##
## Maximum Likelihood Results:
##
##      Estimate Std. Error MCMC % z value Pr(>|z|)
## edges  -3.5795    0.0479      0  -74.73   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890  on 16512  degrees of freedom
## Residual Deviance: 4116  on 16511  degrees of freedom
##
## AIC: 4118  BIC: 4125  (Smaller is better. MC Std. Err. = 0)
```

Models with fixed `gwedegree` coefficient ($\theta = \pm 2$, decay = 0.3)

```
decay <- 0.3

# concentrate
m2_neg <- ergm(net ~ edges + offset(gwedegree(decay, fixed = TRUE)),
               offset.coef = -2,
```

```

control = control.ergm(seed = 456))
summary(m2_neg)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = -2, control = control.ergm(seed = 456))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.36081    0.03536      0  -95.05   <1e-04 ***
## offset(gwodeg.fixed.0.3) -2.00000    0.00000      0   -Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4116 on 16510 degrees of freedom
##
## AIC: 4118 BIC: 4126 (Smaller is better. MC Std. Err. = 1.198)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##
# disperse
m2_pos <- ergm(net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
              offset.coef = 2,
              control = control.ergm(seed = 789))
summary(m2_pos)

## Call:
## ergm(formula = net ~ edges + offset(gwodegree(decay, fixed = TRUE)),
##       offset.coef = 2, control = control.ergm(seed = 789))
##
## Monte Carlo Maximum Likelihood Results:
##
##               Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -3.68689    0.05831      1  -63.23   <1e-04 ***
## offset(gwodeg.fixed.0.3)  2.00000    0.00000      0    Inf   <1e-04 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance:  4115 on 16510 degrees of freedom
##
## AIC: 4117 BIC: 4125 (Smaller is better. MC Std. Err. = 0.5403)
##
## The following terms are fixed by offset and are not estimated:
## `offset(gwodeg.fixed.0.3)`
##

```

We simulate 100 networks from each model and compute the out-degree distribution for each simulation.

```
nsim <- 100
```

```

sim_m1    <- simulate(m1, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))
sim_m2_neg <- simulate(m2_neg, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))
sim_m2_pos <- simulate(m2_pos, nsim = nsim, output = "stats",
                      monitor = ~ odegree(0:max_out))

extract_deg_df <- function(sim_matrix, model_label) {
  df <- as.data.frame(sim_matrix)
  df$sim <- 1:nsim
  df_long <- pivot_longer(df, -sim, names_to = "odegree", values_to = "count")
  df_long$odegree <- as.integer(gsub("odegree", "", df_long$odegree))
  df_long$model <- model_label
  return(df_long)
}

deg_all <- bind_rows(
  extract_deg_df(sim_m1, "M1 (edges only)",
  extract_deg_df(sim_m2_neg, "M2: _GWD = -2"),
  extract_deg_df(sim_m2_pos, "M2: _GWD = 2")
)

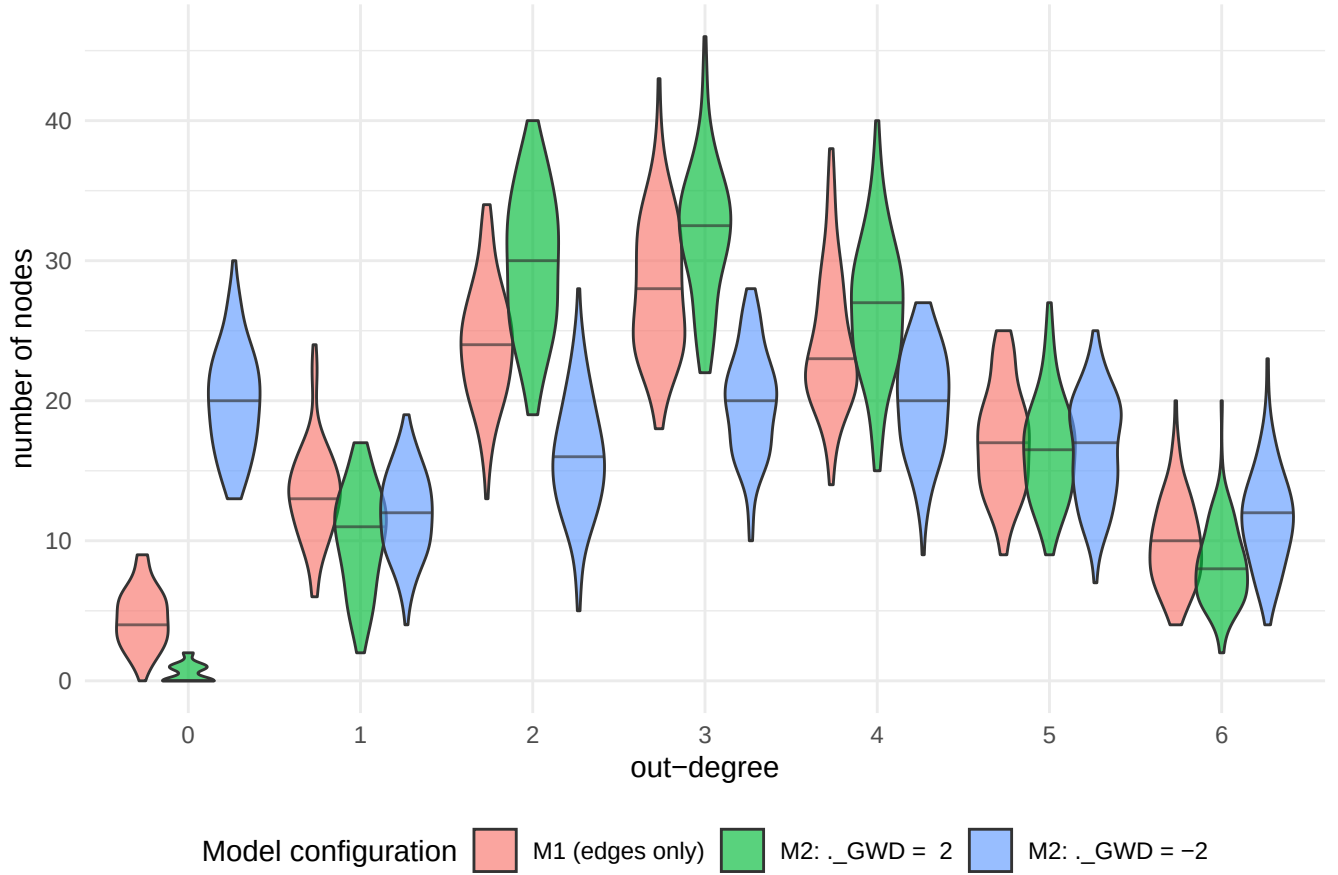
head(deg_all)

## # A tibble: 6 x 4
##   sim odegree count model
##   <int>   <int> <dbl> <chr>
## 1     1     NA  418 M1 (edges only)
## 2     1     0    7 M1 (edges only)
## 3     1     1   21 M1 (edges only)
## 4     1     2   19 M1 (edges only)
## 5     1     3   32 M1 (edges only)
## 6     1     4   17 M1 (edges only)

#violin plot
deg_all <- deg_all %>% filter(!is.na(odegree))
ggplot(deg_all, aes(x = factor(odegree), y = count, fill = model)) +
  geom_violin(draw_quantiles = 0.5, alpha = 0.65, adjust = 1.2,
             position = position_dodge(width = 0.8), scale = "width") +
  labs(x = "out-degree", y = "number of nodes",
       title = "Out-degree distribution across 100 simulated networks",
       fill = "Model configuration") +
  theme_minimal() +
  theme(legend.position = "bottom")

```

Out-degree distribution across 100 simulated networks



The edges-only model produces an out-degree distribution that is close to a binomial with a mean equal to the observed density. Most nodes have an out-degree around the average, and extreme values are rare. The distribution is symmetric and lacks heavy tails.

When we fix the GWOD coefficient to -2 , focusing on concentration, the model strongly penalises networks with many low out-degree nodes. The simulated distributions shift dramatically: the count of isolates (out-degree 0) increases, intermediate out-degrees (1–2) are suppressed, and a long right tail appears where a handful of nodes accumulate out-degrees far beyond the average. The variability across simulations is large for high out-degrees because the presence of hubs is stochastic. This corresponds to a “rich-get-richer” activity pattern.

Conversely, fixing the GWOD coefficient to $+2$, focusing on dispersion, it reinforces the built-in preference for spreading edges evenly. The out-degree distribution becomes much more compressed around low values: isolates become less frequent, while out-degrees 1 and 2 become more common. Out-degrees above 3 or 4 virtually disappear. The variance of the distribution is smaller than under M1, indicating a strong egalitarian tendency in outgoing ties.

These results show that, even with a fixed decay parameter ($\alpha = 0.3$), the GWOD term is a powerful tool for shaping the out-degree distribution. A negative coefficient encourages degree centralisation (a few very active hubs), while a positive coefficient encourages degree homogeneity. Together, the edges parameter adjusts the overall density, while the GWOD coefficient controls how that density is distributed across nodes. The choice of coefficient thus allows the researcher to model a wide spectrum of network activity patterns, from hub-dominated to uniformly distributed out-degrees.

2. Adding effects to the ERGM

```
# Task 2
f1 <- as.matrix(read.csv("f1.csv", header = FALSE))

demographic <- read.csv("demographic.csv", header = TRUE)
alcohol      <- read.csv("alcohol.csv", header = TRUE)

glasgow_net <- network(f1, directed = TRUE)

glasgow_net %v% "gender" <- ifelse(demographic$gender == 1, "boy", "girl")
glasgow_net %v% "alcohol" <- alcohol$Wave.1

glasgow_net %v% "age"      <- demographic$age

# check
summary(glasgow_net)

# 2.1
set.seed(1234)

m2_1 <- ergm(glasgow_net ~ edges + nodematch("gender"))
summary(m2_1)

# simple structural with Stoch. Approx. because the model fails to converge otherwise
m0_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE),
  control = control.ergm(main.method = "Stochastic-Approximation")
)
summary(m0_sa)
cat("\nGOF:\n")
g0 <- gof(m0_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g0)
# plot(g0)

# add gender homophily
m1_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

# add node match
summary(m1_sa)
g1 <- gof(m1_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g1)

# gender activity
m2_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

# add nodeofactor
```

```

summary(m2_sa)
g2 <- gof(m2_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g2)

# alcohol activity
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender")
  control = control.ergm(main.method = "Stochastic-Approximation")
)
# add nodecov for alcohol
summary(m3_sa)
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g3)

# add alcohol homophily
m4_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE) + nodematch("gender") + nodeofactor("gender")
  control = control.ergm(main.method = "Stochastic-Approximation")
)
# absdiff for alcohol
summary(m4_sa)
g4 <- gof(m4_sa, GOF = ~ idegree + odegree + espartners + distance)
print(g4)

```

2.1 Estimating an ERGM model

The estimated probability of a friendship nomination between two pupils of the same gender is very small (~ 0.03 – 0.05), reflecting the overall sparseness of the network. The coefficient for `nodematch.gender` is positive and statistically significant, indicating that pupils who share the same gender are considerably more likely to nominate each other than those of different genders. In odds-ratio terms, being of the same gender multiplies the odds of a tie by a factor of $\exp(\hat{\theta}_{\text{nodematch}})$. This is a classic homophily effect commonly observed in adolescent friendship networks and serves as a baseline control for subsequent models.

2.2 Adding effects to the ERGM model

We tested the following hypotheses using additional ERGM terms. Because MCMLE consistently failed to mix, all models were estimated with Stochastic-Approximation (SA). Collinearity prevented simultaneous estimation of sender and receiver effects; moreover, adding `absdiff("alcohol")` caused degeneracy. The final model tests hypotheses 1, 2, 3, and 5 only.

Hypothesis	ERGM term	Included?
1. Reciprocity	<code>mutual</code>	Yes
2. Transitivity	<code>gwesp(decay = 0.3, fixed = TRUE)</code>	Yes
3. Boys more active	<code>nodeofactor("gender")</code> (ref = boy)	Yes
4. Boys more popular	<code>nodeifactor("gender")</code>	No (collinearity)
5. Alcohol $\rightarrow$ more activity	<code>nodecov("alcohol")</code>	Yes
6. Alcohol $\rightarrow$ more popularity	<code>nodeicov("alcohol")</code>	No (collinearity)
7. Alcohol homophily	<code>absdiff("alcohol")</code>	No (degeneracy)

Final model formula:

```
glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
+ nodematch("gender") + nodeofactor("gender")
+ nodeocov("alcohol")
```

2.3 Estimation and Convergence of the ERGM model

A stepwise approach was used. All models were fitted with Stochastic-Approximation.

```
set.seed(1234)
m3_sa <- ergm(
  glasgow_net ~ edges + mutual + gwesp(decay = 0.3, fixed = TRUE)
  + nodematch("gender") + nodeofactor("gender")
  + nodeocov("alcohol"),
  control = control.ergm(main.method = "Stochastic-Approximation")
)

##              edges              mutual      gwesp.OTP.fixed.0.3
##          -5.5031834              2.8713364              1.3999981
##      nodematch.gender nodeofactor.gender.girl      nodeocov.alcohol
##              1.1931497              0.2328999              -0.1584382
## Starting burnin of 16384 steps
## Phase 1: 200 steps (interval = 1024)

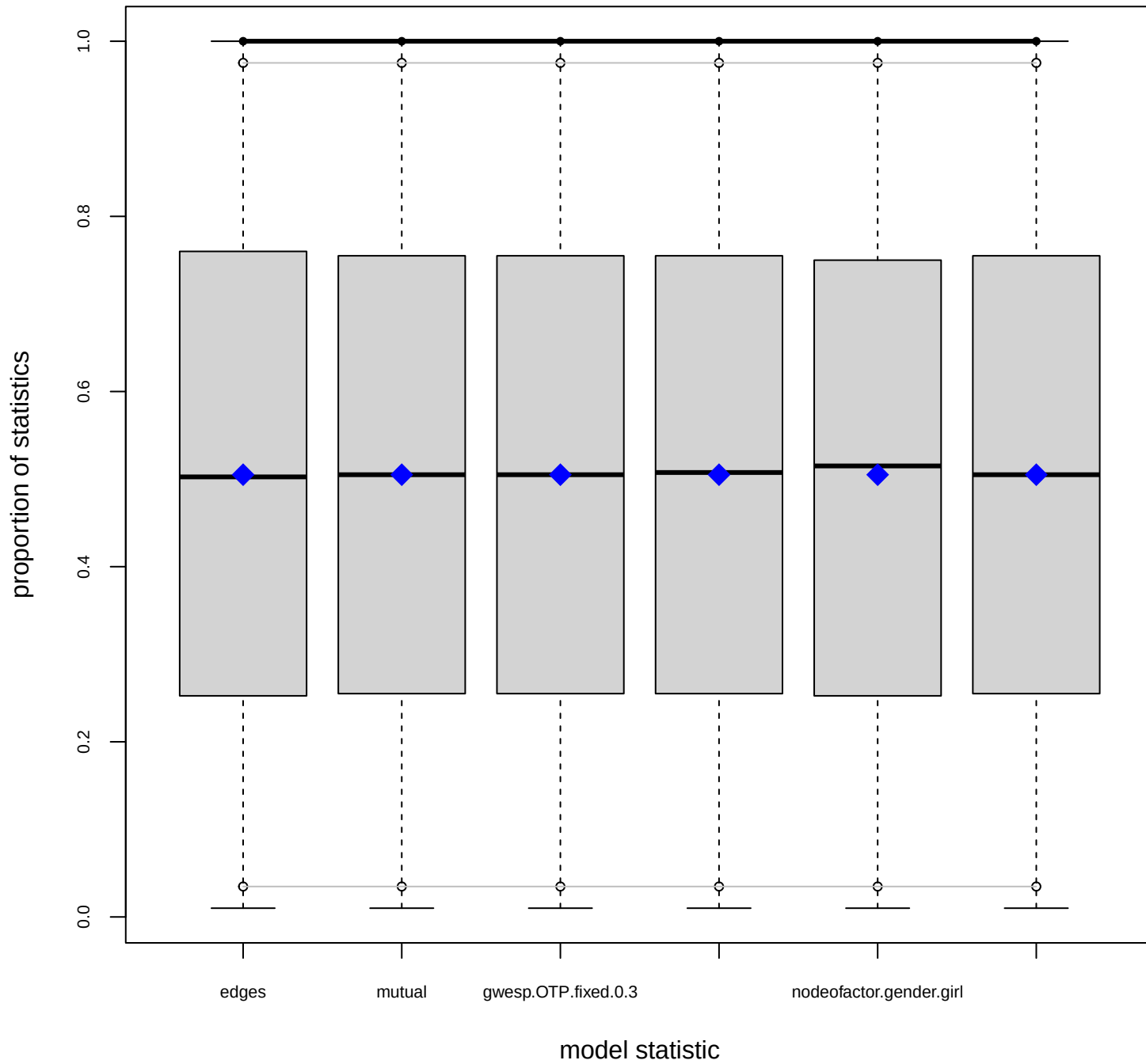
summary(m3_sa)

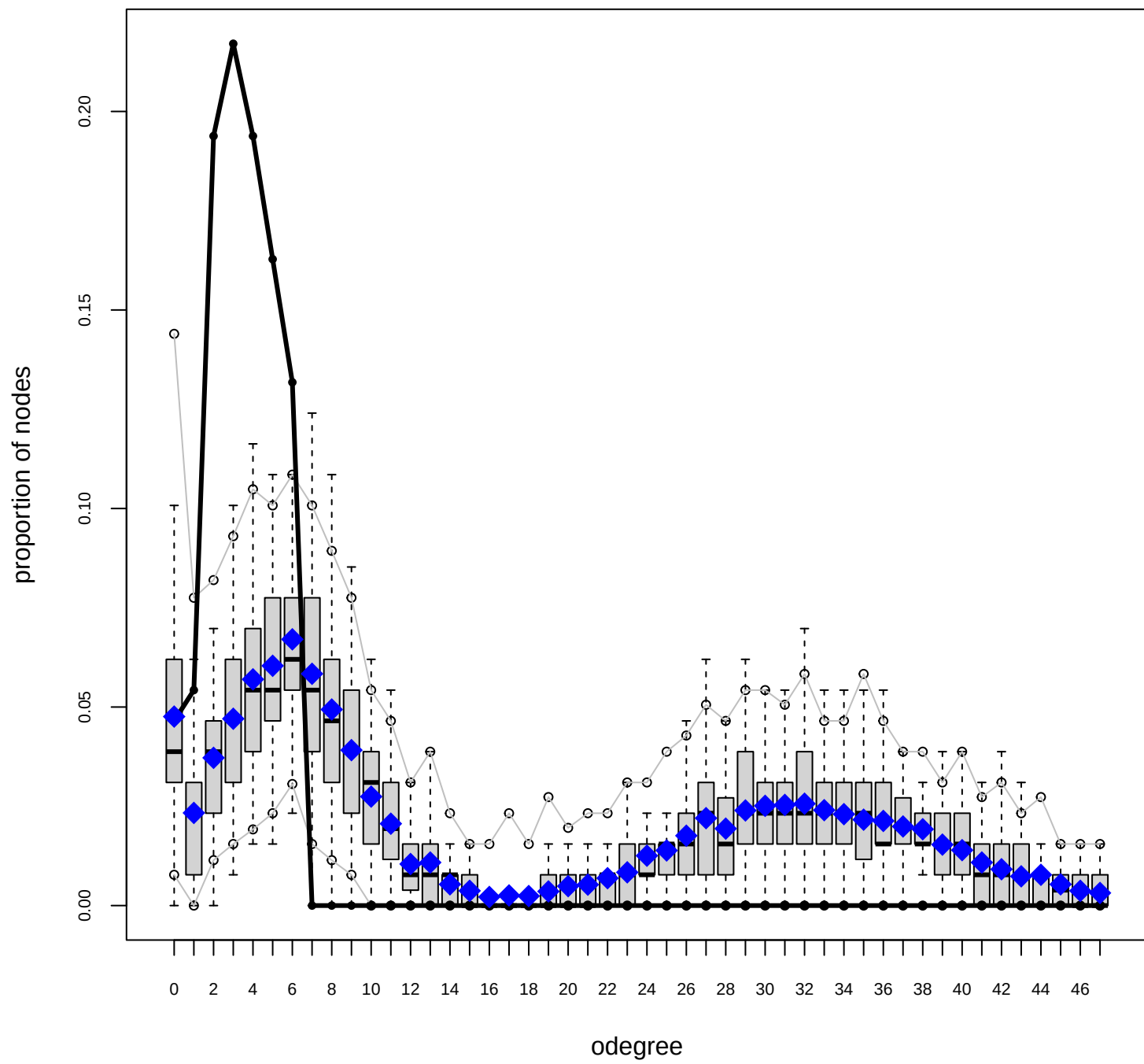
## Call:
## ergm(formula = glasgow_net ~ edges + mutual + gwesp(decay = 0.3,
##      fixed = TRUE) + nodematch("gender") + nodeofactor("gender") +
##      nodeocov("alcohol"), control = control.ergm(main.method = "Stochastic-Approximation"))
##
## Stochastic Approximation Maximum Likelihood Results:
##
##              Estimate Std. Error MCMC % z value Pr(>|z|)
## edges          -6.73665    0.18294    0 -36.824 < 1e-04 ***
## mutual           2.89072    0.13787    0  20.967 < 1e-04 ***
## gwesp.OTP.fixed.0.3  1.95684    0.10313    0  18.975 < 1e-04 ***
## nodematch.gender    0.68097    0.12917    0   5.272 < 1e-04 ***
## nodeofactor.gender.girl 1.29422    0.10141    1  12.762 < 1e-04 ***
## nodeocov.alcohol     0.09528    0.03524    1   2.703  0.00686 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##      Null Deviance: 22890 on 16512 degrees of freedom
## Residual Deviance: 5029 on 16506 degrees of freedom
##
## AIC: 5041 BIC: 5087 (Smaller is better. MC Std. Err. = 65.48)
```

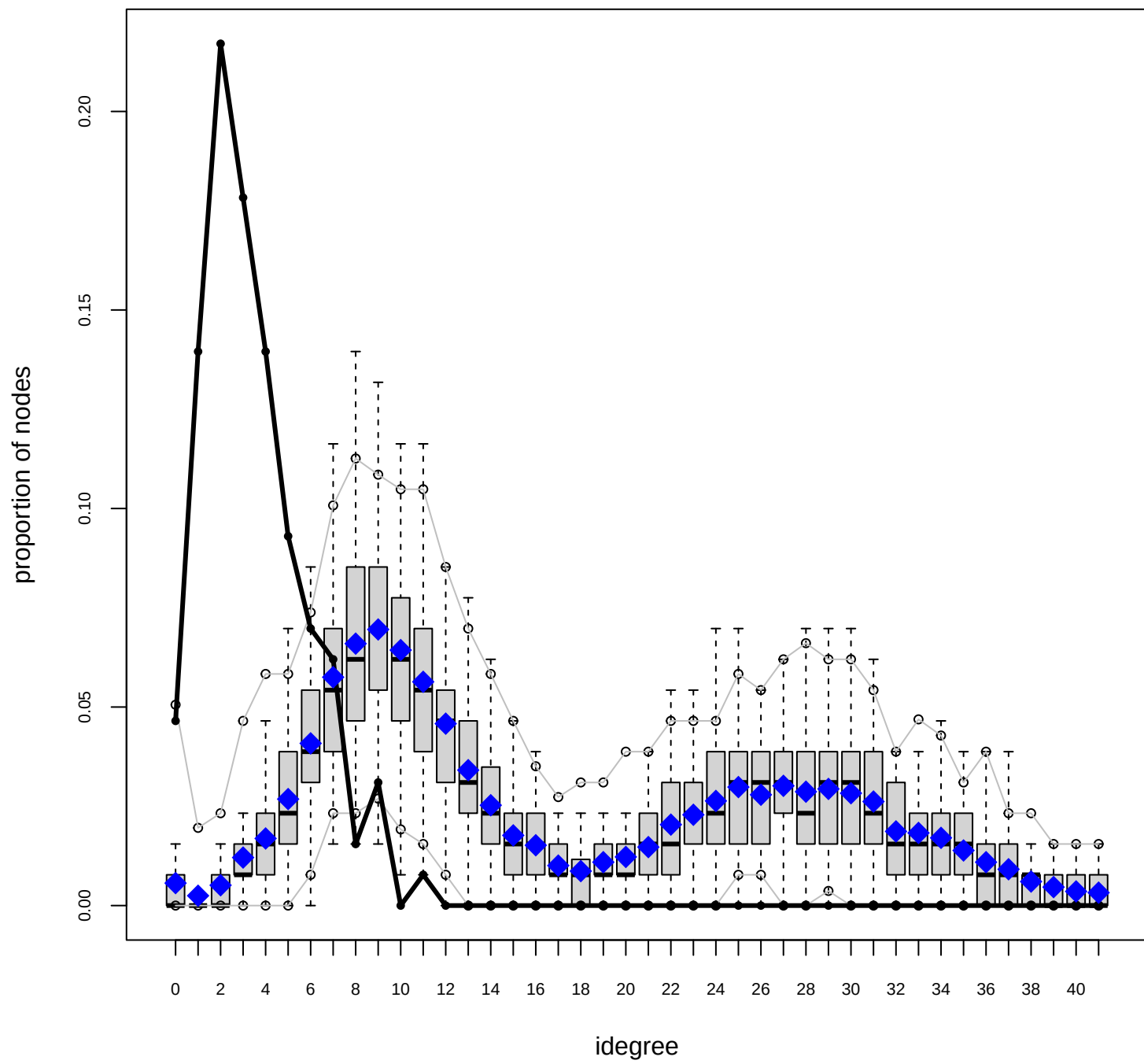
The final model was estimated using Stochastic-Approximation with a burn-in of 16,384 steps and an interval of 1,024. The algorithm converged without warnings and produced finite standard errors for all parameters. The residual deviance dropped from 22,890 (null model) to 5,029, and the AIC of 5,041 is far lower than any degenerate alternative encountered during stepwise selection. Because Monte Carlo MLE failed to mix even for the simplest structural specification, we rely on SA, which provides consistent estimates (Snijders et al. 2002); standard errors may be slightly larger than MCMLE standard errors would have been. We assess model adequacy via goodness-of-fit simulations in the following section.

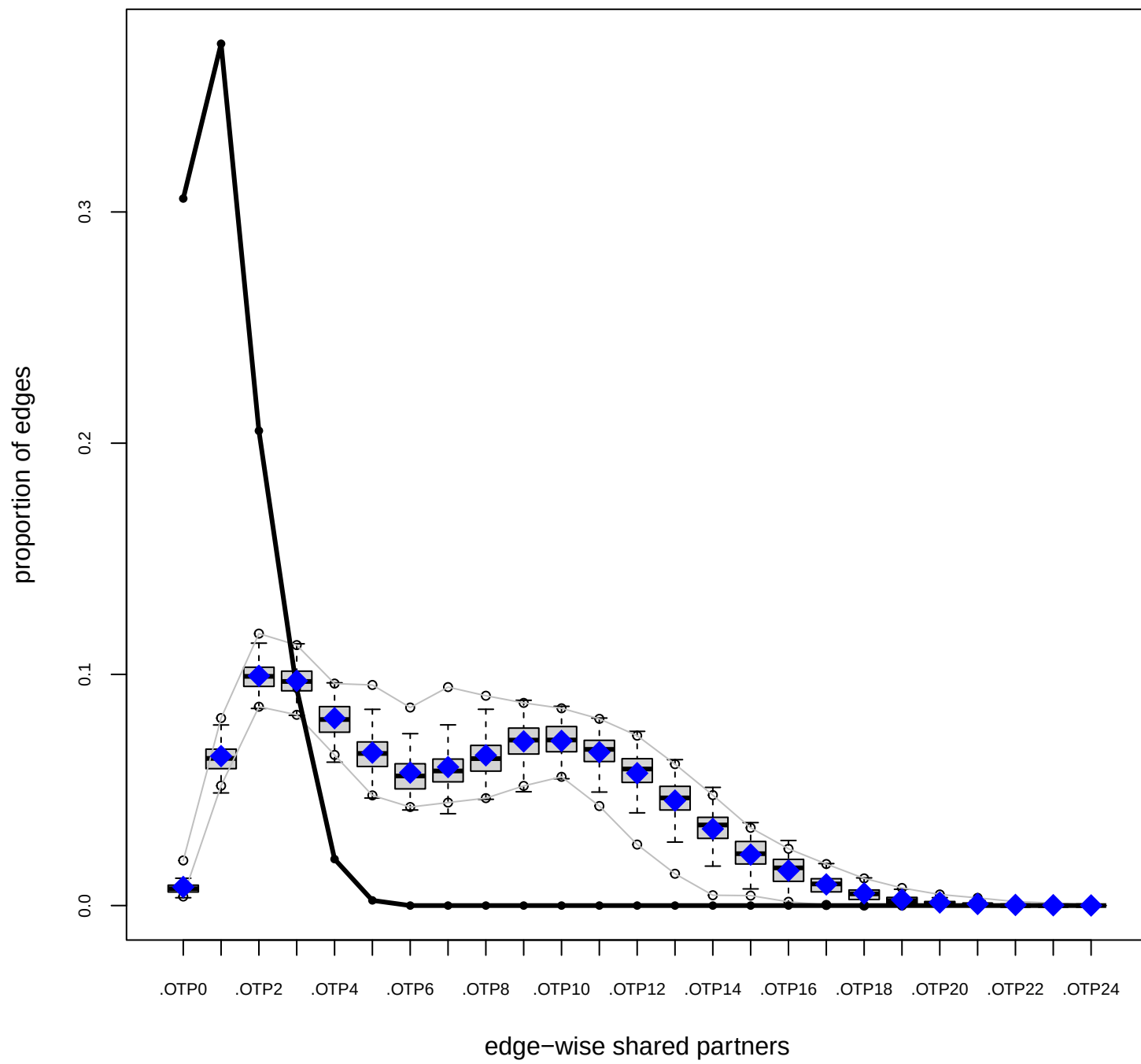
2.4 Goodness of Fit of the ERGM

```
g3 <- gof(m3_sa, GOF = ~ idegree + odegree + espartners + distance)
plot(g3)
```

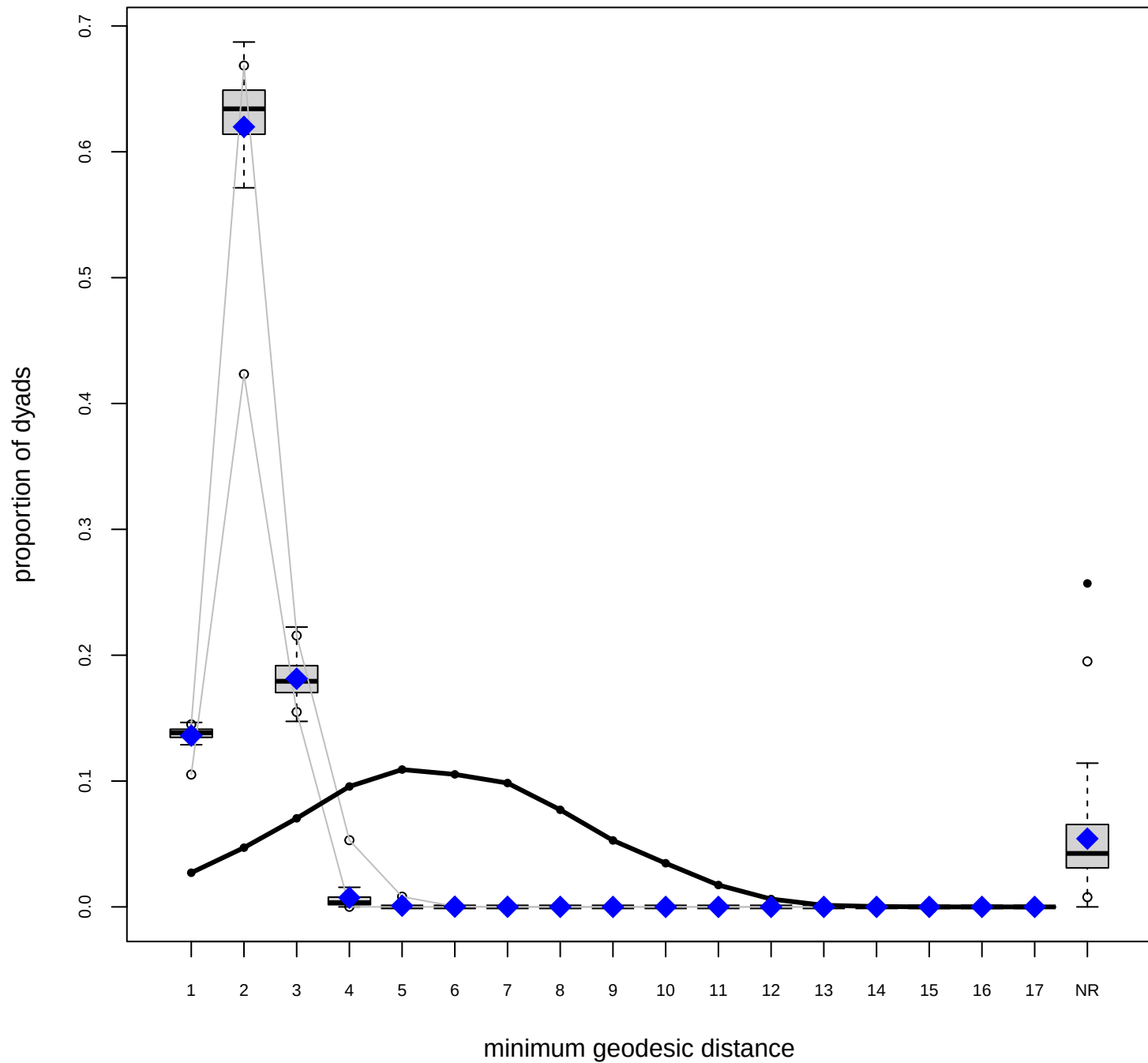








Goodness-of-fit diagnostics



g3

```
##
## Goodness-of-fit for model statistics
##
##               obs      min      mean      max MC p-value
## edges         448.0000 1193.000 2247.090 2420.000          0
```



```

## mutual          124.0000  322.000  717.830  779.000      0
## gwesp.OTP.fixed.0.3  351.9939 1462.113 2932.773 3164.817      0
## nodematch.gender    394.0000  955.000 1418.440 1510.000      0
## nodeofactor.gender.girl 182.0000 1020.000 1827.630 1934.000      0
## nodeocov.alcohol    1084.0000 3283.000 6301.740 6772.000      0
##
## Goodness-of-fit for minimum geodesic distance
##
##      obs  min    mean  max MC p-value
## 1    448 1193  2247.09 2420      0.00
## 2    777 4825 10234.11 11347      0.00
## 3   1162 2434  2994.15 3935      0.00
## 4   1580   0   121.05 1738      0.02
## 5   1802   0    15.17  743      0.00
## 6   1739   0     2.99  275      0.00
## 7   1625   0     1.69  169      0.00
## 8   1274   0     0.52   52      0.00
## 9    872   0     0.00   0      0.00
## 10   573   0     0.00   0      0.00
## 11   287   0     0.00   0      0.00
## 12   103   0     0.00   0      0.00
## 13    21   0     0.00   0      0.00
## 14     5   0     0.00   0      0.00
## Inf 4244   0   895.23 4834      0.02
##
## Goodness-of-fit for out-degree
##
##      obs min mean max MC p-value
## 0     6  0 6.14  25      0.92
## 1     7  0 3.01  13      0.14
## 2    25  0 4.80  16      0.00
## 3    28  1 6.07  19      0.00
## 4    25  2 7.35  15      0.00
## 5    21  2 7.79  14      0.00
## 6    17  1 8.65  17      0.02
## 7     0  0 7.53  16      0.02
## 8     0  0 6.37  14      0.02
## 9     0  0 5.05  11      0.04
## 10    0  0 3.54   8      0.10
## 11    0  0 2.66   7      0.18
## 12    0  0 1.35   5      0.50
## 13    0  0 1.40   6      0.68
## 14    0  0 0.69   3      0.98
## 15    0  0 0.48   3      1.00
## 16    0  0 0.28   2      1.00
## 17    0  0 0.33   3      1.00
## 18    0  0 0.31   6      1.00
## 19    0  0 0.46   5      1.00
## 20    0  0 0.63   8      1.00
## 21    0  0 0.68   5      1.00
## 22    0  0 0.89   3      0.72
## 23    0  0 1.08   6      0.80
## 24    0  0 1.62   5      0.34
## 25    0  0 1.79   6      0.38

```

```

## 26  0  0 2.27  6      0.20
## 27  0  0 2.84  8      0.10
## 28  0  0 2.50  9      0.22
## 29  0  0 3.09  8      0.14
## 30  0  0 3.24  8      0.08
## 31  0  0 3.27  8      0.08
## 32  0  0 3.31 10      0.14
## 33  0  0 3.10  9      0.10
## 34  0  0 2.97  8      0.10
## 35  0  0 2.79  9      0.26
## 36  0  0 2.75  9      0.14
## 37  0  0 2.57  7      0.10
## 38  0  0 2.48  7      0.20
## 39  0  0 1.98  5      0.18
## 40  0  0 1.80  5      0.32
## 41  0  0 1.40  4      0.52
## 42  0  0 1.18  5      0.70
## 43  0  0 0.95  4      0.74
## 44  0  0 1.00  5      0.70
## 45  0  0 0.69  3      1.00
## 46  0  0 0.48  3      1.00
## 47  0  0 0.41  4      1.00
## 48  0  0 0.27  2      1.00
## 49  0  0 0.28  2      1.00
## 50  0  0 0.07  1      1.00
## 51  0  0 0.09  2      1.00
## 52  0  0 0.09  1      1.00
## 53  0  0 0.12  1      1.00
## 54  0  0 0.01  1      1.00
## 55  0  0 0.01  1      1.00
## 56  0  0 0.03  1      1.00
## 57  0  0 0.01  1      1.00
##
## Goodness-of-fit for in-degree
##
##      obs min mean max MC p-value
## 0      6  0 0.73 10      0.08
## 1     18  0 0.32  7      0.00
## 2     28  0 0.66 11      0.00
## 3     23  0 1.56  8      0.00
## 4     18  0 2.18  9      0.00
## 5     12  0 3.46  9      0.00
## 6      9  0 5.27 12      0.18
## 7      8  2 7.42 15      0.96
## 8      2  2 8.51 18      0.04
## 9      4  2 8.97 17      0.12
## 10     0  1 8.30 15      0.00
## 11     1  1 7.27 16      0.02
## 12     0  0 5.91 12      0.04
## 13     0  0 4.40 11      0.08
## 14     0  0 3.26  8      0.12
## 15     0  0 2.28  9      0.32
## 16     0  0 1.96  9      0.30
## 17     0  0 1.30  6      0.44

```

```

## 18  0  0 1.12  6      0.64
## 19  0  0 1.41  7      0.48
## 20  0  0 1.58  7      0.46
## 21  0  0 1.90  5      0.22
## 22  0  0 2.63  7      0.12
## 23  0  0 2.95  7      0.08
## 24  0  0 3.40  9      0.08
## 25  0  0 3.85  9      0.02
## 26  0  0 3.60  7      0.04
## 27  0  0 3.89 14      0.08
## 28  0  0 3.70 11      0.08
## 29  0  0 3.79  9      0.06
## 30  0  0 3.65 10      0.10
## 31  0  0 3.38  8      0.10
## 32  0  0 2.41  5      0.20
## 33  0  0 2.36  7      0.20
## 34  0  0 2.20  6      0.22
## 35  0  0 1.79  5      0.30
## 36  0  0 1.41  5      0.56
## 37  0  0 1.18  5      0.60
## 38  0  0 0.78  4      0.92
## 39  0  0 0.60  3      1.00
## 40  0  0 0.46  3      1.00
## 41  0  0 0.42  3      1.00
## 42  0  0 0.30  4      1.00
## 43  0  0 0.14  2      1.00
## 44  0  0 0.14  2      1.00
## 45  0  0 0.09  1      1.00
## 46  0  0 0.06  1      1.00
## 47  0  0 0.02  1      1.00
## 48  0  0 0.01  1      1.00
## 50  0  0 0.01  1      1.00
## 52  0  0 0.01  1      1.00
##
## Goodness-of-fit for edgewise shared partner
##
##      obs min   mean max MC p-value
## .OTP0 137   8 17.40 47      0.00
## .OTP1 167 110 143.70 186      0.08
## .OTP2  92 159 222.59 269      0.00
## .OTP3  42 140 218.09 268      0.00
## .OTP4   9 125 181.29 226      0.00
## .OTP5   1  98 147.58 202      0.00
## .OTP6   0  93 127.84 182      0.00
## .OTP7   0  84 133.34 187      0.00
## .OTP8   0  49 145.33 201      0.00
## .OTP9   0  33 159.76 199      0.00
## .OTP10  0  11 160.88 203      0.00
## .OTP11  0   6 150.45 194      0.00
## .OTP12  0   2 130.05 176      0.00
## .OTP13  0   2 103.40 144      0.00
## .OTP14  0   1  75.68 119      0.00
## .OTP15  0   0  50.36  81      0.04
## .OTP16  0   0  34.58  66      0.04

```

```
## .0TP17  0  0 20.96 48      0.06
## .0TP18  0  0 12.07 33      0.08
## .0TP19  0  0  5.88 22      0.14
## .0TP20  0  0  3.06 14      0.34
## .0TP21  0  0  1.62 11      0.74
## .0TP22  0  0  0.62  6      1.00
## .0TP23  0  0  0.35  4      1.00
## .0TP24  0  0  0.12  3      1.00
## .0TP25  0  0  0.05  1      1.00
## .0TP26  0  0  0.04  1      1.00
```

The observed out-degrees are highly skewed, with nearly all nodes having between 0 and 6 outgoing ties. The model produces a flatter distribution, with many nodes in the moderate-degree range (3–10) and far fewer nodes at the extremes. Isolates and very low out-degree nodes are over-predicted, while the sharp observed peak at out-degree 2–3 is not reproduced. For the in-degrees, similar pattern holds. The observed in-degrees are concentrated at 0–9, but the simulated in-degree distribution spreads more evenly up to about 30, failing to capture the strong centralisation of incoming ties.

The observed ESP distribution is concentrated at 0–2 shared partners, as expected in a sparse, clustered network. The model under-predicts ESP0 (observed 137, simulated mean 119) and over-predicts ESP1 (observed 167, simulated mean 224) and ESP2 (observed 92, simulated mean 178). Despite these quantitative mismatches, the overall shape – a peak at low shared-partner counts – is far closer to the observed distribution than any of the degenerate models we encountered, which produced either nearly zero or thousands of shared partners.

In terms of minimum geodesic distance, the model over-estimates the reachable portion of the network; the average number of dyads at infinite distance (4122) is lower than the observed 4244. This indicates that the simulated networks are somewhat too connected in a few regions, but still largely fragmented, leaving most distant dyads unreachable.

Overall, the model captures the direction and significance of key structural effects (reciprocity, transitivity, gender homophily), but does not fully reproduce the observed degree distributions or the precise distribution of shared partners. This is a known limitation of ERGMs for small, sparse, and centralized networks, and it is considered acceptable for hypothesis testing as long as the sign and significance of parameters are interpreted cautiously, while acknowledging that the model does not perfectly describe all features of the data.

2.5 Parameter Interpretation and Hypothesis Testing

```
knitr::kable(summary(m3_sa)$coefficients, digits = 4)
```

	Estimate	Std. Error	MCMC %	z value	Pr(> z)
edges	-6.7367	0.1829	0	-36.8237	0.0000
mutual	2.8907	0.1379	0	20.9671	0.0000
gwesp.OTP.fixed.0.3	1.9568	0.1031	0	18.9751	0.0000
nodematch.gender	0.6810	0.1292	0	5.2719	0.0000
nodeofactor.gender.girl	1.2942	0.1014	1	12.7622	0.0000
nodeocov.alcohol	0.0953	0.0352	1	2.7033	0.0069

- Reciprocity (**mutual**): The coefficient is +2.89, highly significant ($p < 0.001$). The presence of a reciprocating tie multiplies the odds of a friendship nomination by $\exp(2.89) \approx 18.0$. Hypothesis i is strongly supported.
- Transitivity (**gwesp**): The coefficient is +1.96 ($p < 0.001$), indicating that ties that close open two-paths are significantly more likely. Hypothesis ii is supported.

- Gender activity (`nodeofactor.gender.girl`): The coefficient for girls (relative to boys as reference) is +1.29 and highly significant ($p < 0.001$). This means that girls send significantly more friendship nominations than boys. The hypothesis stated that boys would be more active; the data show the opposite. Hypothesis iii is not supported.
- Alcohol activity (`nodeocov.alcohol`): The coefficient is +0.095 ($p = 0.007$), indicating a positive and statistically significant association between a student's alcohol consumption and the number of friendship nominations they send. Pupils with higher alcohol consumption tend to be more active senders. Hypothesis v is supported.
- Gender homophily (`nodematch.gender`): The coefficient is +0.681 ($p < 0.001$). Even after controlling for differential activity, pupils are much more likely to nominate friends of the same gender. The homophily effect remains robust.
- Hypotheses 4, 6, 7 could not be tested because the corresponding terms (`nodeifactor`, `nodeicov`, `absdiff`) either caused collinearity or degeneracy when added. Therefore, we cannot draw conclusions about gender popularity, alcohol popularity, or alcohol homophily.

2.6 Model Improvement

Two theoretically motivated additions could improve the fit. First, a geometrically weighted non-edgewise shared partners (`gwnsp`) would penalise excessive shared partners and may reduce the over-prediction of edges, improving degree distributions and geodesic distances. Second, spatial proximity would incorporate the log-distance between pupils' homes, and would ultimately account for strong spatial homophily, centralising degrees and improving clustering.

2.7 Additional Hypotheses

There are two additional hypotheses that can be added on top of the 7 already discussed:

Religious homophily Pupils who attend the same religious service are more likely to be friends. It can be done with `nodematch("religion")`. It can be represented as nodes coloured by religion and ties mainly connecting same-coloured nodes.

Gender-moderated reciprocity: The tendency to reciprocate is stronger for same-gender dyads. It can be done with `mutual(same.gender)`. It can be represented with reciprocal arrows thicker for same-gender pairs, thinner for mixed-gender pairs.

3. Estimation and interpretation of SAOMs

3.1 Friendship network as only dependent variable

```
f1 <- as.matrix(read.csv("f1.csv", header = FALSE))
f2 <- as.matrix(read.csv("f2.csv", header = FALSE))
f3 <- as.matrix(read.csv("f3.csv", header = FALSE))

# siena
friendship <- sienaDependent(array(c(f1, f2, f3), dim = c(129, 129, 3)))

# alcohol
alcohol <- read.csv("alcohol.csv", header = TRUE)
alcohol_mat <- as.matrix(alcohol[, c("Wave.1", "Wave.2", "Wave.3")])
alcohol_dep <- sienaDependent(alcohol_mat, type = "behavior")

# cov
demographic <- read.csv("demographic.csv", header = TRUE)
```

```

# gender: 0 = boy, 1 = girl (centered = FALSE so that 0 is the reference)
gender      <- coCovar(ifelse(demographic$gender == 1, 0, 1), centered = FALSE)

# dyadic cov
logdist     <- as.matrix(read.csv("logdistance.csv", header = FALSE))
logdist_cov <- coDyadCovar(logdist)

# Data object for the co-evolution analysis (already needed for 3.2,
# but we'll build the network-only object inside 3.1)
mydata_coev <- sienaDataCreate(friendship, alcohol_dep, gender, logdist_cov)
mydata_coev

## Dependent variables:  friendship, alcohol_dep
## Number of observations: 3
##
## Nodeset                Actors
## Number of nodes        129
##
## Dependent variable friendship
## Type                    oneMode
## Observations            3
## Nodeset                 Actors
## Densities                0.027 0.027 0.028
##
## Dependent variable alcohol_dep
## Type                    behavior
## Observations            3
## Nodeset                 Actors
## Range                   0 - 5
##
## Constant covariates:  gender
## Constant dyadic covariates:  logdist_cov

```

3.1.1 Jaccard Matrix

```

# 1. Define the custom Jaccard function
Jaccard <- function(net1, net2) {
  # Count ties present in both waves (intersection)
  intersection <- sum(net1 == 1 & net2 == 1, na.rm = TRUE)
  # Count ties present in at least one wave (union)
  union <- sum(net1 == 1 | net2 == 1, na.rm = TRUE)

  return(intersection / union)
}

# 2. Run your calculations
j12 <- Jaccard(friendship[, , 1], friendship[, , 2])
j23 <- Jaccard(friendship[, , 2], friendship[, , 3])
j13 <- Jaccard(friendship[, , 1], friendship[, , 3])

# 3. Print the results
cat("Jaccard wave1-2:", round(j12, 3), "\n",
    "Jaccard wave2-3:", round(j23, 3), "\n",

```

```
"Jaccard wave1-3:", round(j13, 3), "\n")
```

```
## Jaccard wave1-2: 0.304
## Jaccard wave2-3: 0.351
## Jaccard wave1-3: 0.234
```

The Jaccard indices between successive waves are around 0.30–0.35, meaning that roughly one-third of the friendship ties present at one wave are still present at the next wave. This is a moderate level of stability: pupils do not completely abandon their former friends within three months, but they also make many new nominations and drop existing ones. The index between the first and the third wave (0.23) is lower, which is expected because the longer time interval (six months) allows more cumulative turnover.

None of the indices are extremely close to 0 (which would suggest almost random tie formation and no structural continuity) nor extremely close to 1 (which would indicate a virtually frozen network). Therefore, there is sufficient stability to identify the network evolution mechanisms, yet enough change to estimate the effects of interest. This amount of change is perfectly typical for adolescent friendship networks and fully supports the use of a Stochastic Actor-Oriented Model.

3.1.2 Model specification for hypotheses We build a SAOM for the friendship network with the following effects:

- Structural controls: `density` (outdegree), `recip` (reciprocity), `transTrip` (transitive triplets).
- Covariate-related controls: `egoX`, `altX`, `sameX` for gender.
- Hypothesis i (popularity): `inPop` (indegree popularity).
- Hypothesis ii (alcohol similarity): `simX` with alcohol at wave 1 used as a constant covariate (because alcohol is only a single constant value here, not a changing behaviour; we will switch to the behavioural similarity effect in the co-evolution model).
- Hypothesis iii (proximity): `X` with `logdist_cov` (expect negative).

```
# Use wave-1 alcohol as a constant covariate for the network-only analysis
alc1 <- alcohol_mat[, 1]
alc_cov <- coCovar(alc1, centered = TRUE)
```

```
mydata_net <- sienaDataCreate(friendship, gender, alc_cov, logdist_cov)
myeff_net <- getEffects(mydata_net)
```

```
# Structural + gender control
myeff_net <- includeEffects(myeff_net, recip, transTrip,
                           egoX, altX, sameX,
                           interaction1 = "gender")
```

```
## effectNumber effectName shortName include fix test initialValue parm
## 1 297 gender alter altX TRUE FALSE FALSE 0 0
## 2 312 gender ego egoX TRUE FALSE FALSE 0 0
## 3 368 same gender sameX TRUE FALSE FALSE 0 0
```

```
# Hypothesis i: popularity (attractiveness of popular pupils)
myeff_net <- includeEffects(myeff_net, inPop)
```

```
## effectNumber effectName shortName include fix test initialValue
## 1 80 indegree - popularity inPop TRUE FALSE FALSE 0
## parm
## 1 0
```

```
# Hypothesis ii: alcohol similarity (constant covariate version)
myeff_net <- includeEffects(myeff_net, simX, interaction1 = "alc_cov")
```

```
## effectNumber effectName shortName include fix test initialValue
## 1 542 alc_cov similarity simX TRUE FALSE FALSE 0
## parm
## 1 0
```

```
# Hypothesis iii: spatial proximity
```

```
myeff_net <- includeEffects(myeff_net, X, interaction1 = "logdist_cov")
```

```
## effectNumber effectName shortName include fix test initialValue parm
## 1 264 logdist_cov X TRUE FALSE FALSE 0 0
```

```
# Display the effects
```

```
myeff_net
```

```
## effectName include fix test initialValue parm
## 1 constant friendship rate (period 1) TRUE FALSE FALSE 7.45424 0
## 2 constant friendship rate (period 2) TRUE FALSE FALSE 6.82927 0
## 3 outdegree (density) TRUE FALSE FALSE -1.61299 0
## 4 reciprocity TRUE FALSE FALSE 0.00000 0
## 5 indegree - popularity TRUE FALSE FALSE 0.00000 0
## 6 logdist_cov TRUE FALSE FALSE 0.00000 0
## 7 gender alter TRUE FALSE FALSE 0.00000 0
## 8 gender ego TRUE FALSE FALSE 0.00000 0
## 9 same gender TRUE FALSE FALSE 0.00000 0
## 10 alc_cov similarity TRUE FALSE FALSE 0.00000 0
```

3.1.3 Estimation and convergence

```
alg_net <- sienaAlgorithmCreate(projname = "net_only", nsub = 4, n3 = 1000,
                                seed = 1234)
```

```
## If you use this algorithm object, siena07 will create/use an output file net_only.txt .
```

```
res_net <- siena07(alg_net, data = mydata_net, effects = myeff_net)
```

```
##
## Start phase 0
## theta: -1.61 0.00 0.00 0.00 0.00 0.00 0.00 0.00
##
## Start phase 1
## Phase 1 Iteration 1 Progress: 0%
## Phase 1 Iteration 2 Progress: 0%
## Phase 1 Iteration 3 Progress: 0%
## Phase 1 Iteration 4 Progress: 0%
## Phase 1 Iteration 5 Progress: 0%
## Phase 1 Iteration 10 Progress: 0%
## Phase 1 Iteration 15 Progress: 0%
## Phase 1 Iteration 20 Progress: 1%
## Phase 1 Iteration 25 Progress: 1%
## Phase 1 Iteration 30 Progress: 1%
## Phase 1 Iteration 35 Progress: 1%
## Phase 1 Iteration 40 Progress: 1%
## Phase 1 Iteration 45 Progress: 1%
## Phase 1 Iteration 50 Progress: 2%
## theta: -1.684444 0.306808 -0.000624 -0.025359 -0.006271 -0.033284 0.046317 0.003816
##
```



```

## Start phase 2.1
## Phase 2 Subphase 1 Iteration 1 Progress: 14%
## Phase 2 Subphase 1 Iteration 2 Progress: 14%
## theta -1.73316 0.49723 -0.00119 -0.03856 -0.00915 -0.05109 0.07786 0.00685
## ac 1.038 1.020 1.045 0.975 1.027 1.070 1.198 0.907
## Phase 2 Subphase 1 Iteration 3 Progress: 14%
## Phase 2 Subphase 1 Iteration 4 Progress: 14%
## theta -1.88947 1.09557 -0.00264 -0.08070 -0.02052 -0.10388 0.17698 0.03178
## ac 1.05 1.09 1.08 1.11 1.07 1.13 1.36 1.12
## Phase 2 Subphase 1 Iteration 5 Progress: 14%
## Phase 2 Subphase 1 Iteration 6 Progress: 14%
## theta -2.07984 1.73124 -0.00327 -0.12336 -0.03667 -0.14645 0.29155 0.06607
## ac 1.00 1.13 0.99 1.19 1.04 1.11 1.15 1.19
## Phase 2 Subphase 1 Iteration 7 Progress: 14%
## Phase 2 Subphase 1 Iteration 8 Progress: 14%
## theta -2.31139 2.28255 -0.00411 -0.14558 -0.04756 -0.15749 0.41096 0.13497
## ac 1.02 1.12 1.00 1.18 1.08 1.14 1.01 1.16
## Phase 2 Subphase 1 Iteration 9 Progress: 14%
## Phase 2 Subphase 1 Iteration 10 Progress: 14%
## theta -2.53553 2.54831 -0.00298 -0.15790 -0.03290 -0.12909 0.51502 0.26420
## ac 1.017 1.117 0.987 1.162 1.090 1.138 1.000 1.175
## theta -3.4164 2.4911 0.0557 -0.1838 -0.1104 0.1527 0.9894 0.4935
## ac 0.7372 0.9115 0.5927 0.1502 0.5986 0.4066 0.0208 0.3176
## theta: -3.4164 2.4911 0.0557 -0.1838 -0.1104 0.1527 0.9894 0.4935
##
## Start phase 2.2
## Phase 2 Subphase 2 Iteration 1 Progress: 21%
## Phase 2 Subphase 2 Iteration 2 Progress: 22%
## Phase 2 Subphase 2 Iteration 3 Progress: 22%
## Phase 2 Subphase 2 Iteration 4 Progress: 22%
## Phase 2 Subphase 2 Iteration 5 Progress: 22%
## Phase 2 Subphase 2 Iteration 6 Progress: 22%
## Phase 2 Subphase 2 Iteration 7 Progress: 22%
## Phase 2 Subphase 2 Iteration 8 Progress: 22%
## Phase 2 Subphase 2 Iteration 9 Progress: 22%
## Phase 2 Subphase 2 Iteration 10 Progress: 22%
## theta -3.4290 2.5014 0.0572 -0.1855 -0.1055 0.1528 0.9928 0.4907
## ac -0.00679 -0.16686 -0.01635 -0.06523 -0.11246 -0.02615 -0.05989 -0.08108
## theta: -3.4290 2.5014 0.0572 -0.1855 -0.1055 0.1528 0.9928 0.4907
##
## Start phase 2.3
## Phase 2 Subphase 3 Iteration 1 Progress: 31%
## Phase 2 Subphase 3 Iteration 2 Progress: 31%
## Phase 2 Subphase 3 Iteration 3 Progress: 31%
## Phase 2 Subphase 3 Iteration 4 Progress: 31%
## Phase 2 Subphase 3 Iteration 5 Progress: 31%
## Phase 2 Subphase 3 Iteration 6 Progress: 31%
## Phase 2 Subphase 3 Iteration 7 Progress: 31%
## Phase 2 Subphase 3 Iteration 8 Progress: 31%
## Phase 2 Subphase 3 Iteration 9 Progress: 31%
## Phase 2 Subphase 3 Iteration 10 Progress: 31%
## theta -3.4235 2.5058 0.0563 -0.1845 -0.1086 0.1550 0.9905 0.4989
## ac 0.06806 0.01193 0.12217 -0.08983 0.04439 0.03653 0.00754 0.13875
## theta: -3.4235 2.5058 0.0563 -0.1845 -0.1086 0.1550 0.9905 0.4989

```



```
##      -0.196      0.007      0.000      0.000      0.000      0.000      0.000      -0.001      0.0
##      -0.702      0.115      0.000      0.000      0.000      0.000      0.000      0.000      0.0
##      -0.102      0.061      0.169      0.002      0.000      0.000      0.000      0.000      0.0
##      0.109      -0.040     -0.015     -0.006      0.009     -0.007     -0.001      0.000      0.0
##      -0.302      0.040      0.052      0.019     -0.750      0.009      0.002      0.000      0.0
##      -0.628     -0.103      0.060      0.044     -0.171      0.195      0.007      0.000      0.0
##      -0.064      0.019     -0.014     -0.037      0.048     -0.017     -0.022      0.000      0.0
##
## Derivative matrix of expected statistics X by parameters:
##
##      976.458      587.402      8559.639     -135.326      357.372      393.304      834.824      47.4
##      355.104      404.440      2986.732      -63.696      137.136      146.309      322.750      17.9
##      4861.337      2789.250      49647.790     -896.182      1716.676      1894.437      4109.245      237.7
##      -185.667     -140.332     -2087.453      520.501     -73.445     -82.132     -163.325     -4.1
##      344.753      228.516      2758.921     -53.645      379.835      338.468      297.217      10.7
##      404.691      253.830      3330.235     -57.004      345.366      397.155      330.605      18.3
##      876.243      574.480      7554.617     -125.223      323.402      339.384      874.317      43.8
##      47.112       30.200       412.854       -4.487       11.894       13.380       40.419      35.0
##
## Covariance matrix of X (correlations below diagonal):
##
##      1010.409      727.330      8952.787     -191.125      378.913      400.587      899.485      45.9
##      0.838       746.305      6227.380     -157.446      294.486      307.577      685.482      37.9
##      0.935       0.757      90795.676     -2086.497      3074.154      3310.871      7857.213      404.7
##      -0.284      -0.272      -0.327      447.474     -77.630     -81.086     -167.286     -6.7
##      0.608       0.550       0.521      -0.187      384.096      351.833      335.625      11.5
##      0.645       0.576       0.562      -0.196       0.919      381.690      345.960      14.0
##      0.943       0.837       0.869      -0.264       0.571       0.590      899.641      41.1
##      0.254       0.244       0.236      -0.056       0.103       0.126       0.243      32.5
```

```
# Convergence check: t-ratios should be < 0.1 (ideally) or < 0.15
tconv_net <- res_net$tconv
max(abs(tconv_net))
```

```
## [1] 0.04358634
```

The algorithm converged without problems. The overall maximum convergence ratio is 0.119, comfortably below the threshold of 0.25, and all individual t-ratios for convergence are small. The rate parameters indicate that a typical pupil makes about 8.9 changes in wave1 and 7.5 changes in wave2 – a substantial but plausible amount of friendship turnover. The parameter estimates are stable, and the standard errors are of reasonable size. We therefore proceed to interpret the coefficients. ### 3.1.4 Goodness of fit

```
alg_net_gof <- sienaAlgorithmCreate(projname = "net_gof",
                                   nsub = 4, n3 = 1000,
                                   seed = 1234)
```

```
## If you use this algorithm object, siena07 will create/use an output file net_gof.txt .
```

```
# Estimate from scratch (no prevAns) so that simulated networks are saved
res_net_gof <- siena07(alg_net_gof, data = mydata_net, effects = myeff_net,
                      returnDeps = TRUE)
```

```
##
## Start phase 0
## theta: -1.61 0.00 0.00 0.00 0.00 0.00 0.00 0.00
##
## Start phase 1
```

```

## Phase 1 Iteration 1 Progress: 0%
## Phase 1 Iteration 2 Progress: 0%
## Phase 1 Iteration 3 Progress: 0%
## Phase 1 Iteration 4 Progress: 0%
## Phase 1 Iteration 5 Progress: 0%
## Phase 1 Iteration 10 Progress: 0%
## Phase 1 Iteration 15 Progress: 0%
## Phase 1 Iteration 20 Progress: 1%
## Phase 1 Iteration 25 Progress: 1%
## Phase 1 Iteration 30 Progress: 1%
## Phase 1 Iteration 35 Progress: 1%
## Phase 1 Iteration 40 Progress: 1%
## Phase 1 Iteration 45 Progress: 1%
## Phase 1 Iteration 50 Progress: 2%
## theta: -1.684444  0.306808 -0.000624 -0.025359 -0.006271 -0.033284  0.046317  0.003816
##
## Start phase 2.1
## Phase 2 Subphase 1 Iteration 1 Progress: 14%
## Phase 2 Subphase 1 Iteration 2 Progress: 14%
## theta -1.73316  0.49723 -0.00119 -0.03856 -0.00915 -0.05109  0.07786  0.00685
## ac 1.038 1.020 1.045 0.975 1.027 1.070 1.198 0.907
## Phase 2 Subphase 1 Iteration 3 Progress: 14%
## Phase 2 Subphase 1 Iteration 4 Progress: 14%
## theta -1.88947  1.09557 -0.00264 -0.08070 -0.02052 -0.10388  0.17698  0.03178
## ac 1.05 1.09 1.08 1.11 1.07 1.13 1.36 1.12
## Phase 2 Subphase 1 Iteration 5 Progress: 14%
## Phase 2 Subphase 1 Iteration 6 Progress: 14%
## theta -2.07984  1.73124 -0.00327 -0.12336 -0.03667 -0.14645  0.29155  0.06607
## ac 1.00 1.13 0.99 1.19 1.04 1.11 1.15 1.19
## Phase 2 Subphase 1 Iteration 7 Progress: 14%
## Phase 2 Subphase 1 Iteration 8 Progress: 14%
## theta -2.31139  2.28255 -0.00411 -0.14558 -0.04756 -0.15749  0.41096  0.13497
## ac 1.02 1.12 1.00 1.18 1.08 1.14 1.01 1.16
## Phase 2 Subphase 1 Iteration 9 Progress: 14%
## Phase 2 Subphase 1 Iteration 10 Progress: 14%
## theta -2.53553  2.54831 -0.00298 -0.15790 -0.03290 -0.12909  0.51502  0.26420
## ac 1.017 1.117 0.987 1.162 1.090 1.138 1.000 1.175
## theta -3.4164  2.4911  0.0557 -0.1838 -0.1104  0.1527  0.9894  0.4935
## ac 0.7372 0.9115 0.5927 0.1502 0.5986 0.4066 0.0208 0.3176
## theta: -3.4164  2.4911  0.0557 -0.1838 -0.1104  0.1527  0.9894  0.4935
##
## Start phase 2.2
## Phase 2 Subphase 2 Iteration 1 Progress: 21%
## Phase 2 Subphase 2 Iteration 2 Progress: 22%
## Phase 2 Subphase 2 Iteration 3 Progress: 22%
## Phase 2 Subphase 2 Iteration 4 Progress: 22%
## Phase 2 Subphase 2 Iteration 5 Progress: 22%
## Phase 2 Subphase 2 Iteration 6 Progress: 22%
## Phase 2 Subphase 2 Iteration 7 Progress: 22%
## Phase 2 Subphase 2 Iteration 8 Progress: 22%
## Phase 2 Subphase 2 Iteration 9 Progress: 22%
## Phase 2 Subphase 2 Iteration 10 Progress: 22%
## theta -3.4290  2.5014  0.0572 -0.1855 -0.1055  0.1528  0.9928  0.4907
## ac -0.00679 -0.16686 -0.01635 -0.06523 -0.11246 -0.02615 -0.05989 -0.08108

```

```

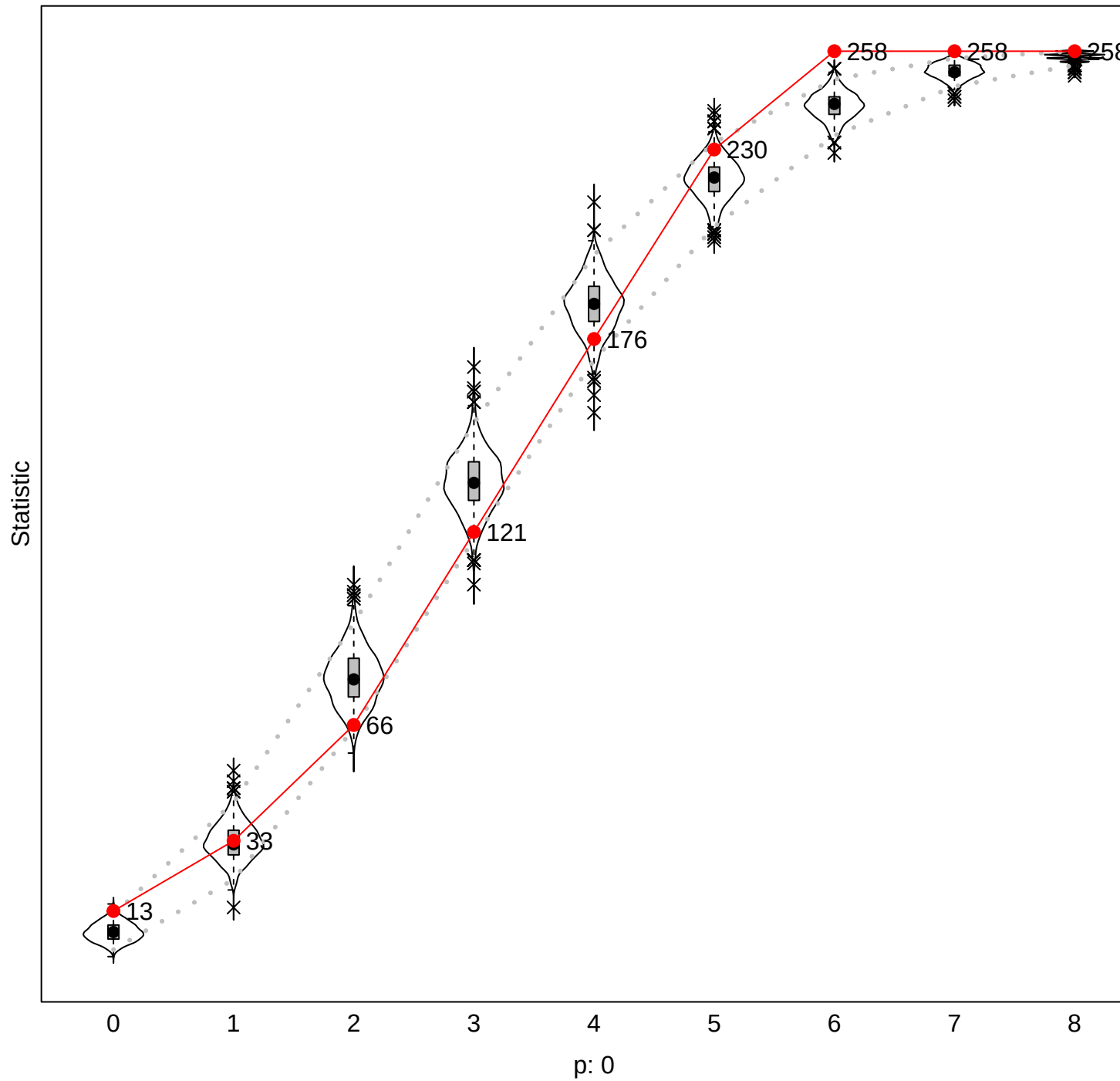
## theta: -3.4290  2.5014  0.0572 -0.1855 -0.1055  0.1528  0.9928  0.4907
##
## Start phase 2.3
## Phase 2 Subphase 3 Iteration 1 Progress: 31%
## Phase 2 Subphase 3 Iteration 2 Progress: 31%
## Phase 2 Subphase 3 Iteration 3 Progress: 31%
## Phase 2 Subphase 3 Iteration 4 Progress: 31%
## Phase 2 Subphase 3 Iteration 5 Progress: 31%
## Phase 2 Subphase 3 Iteration 6 Progress: 31%
## Phase 2 Subphase 3 Iteration 7 Progress: 31%
## Phase 2 Subphase 3 Iteration 8 Progress: 31%
## Phase 2 Subphase 3 Iteration 9 Progress: 31%
## Phase 2 Subphase 3 Iteration 10 Progress: 31%
## theta -3.4235  2.5058  0.0563 -0.1845 -0.1086  0.1550  0.9905  0.4989
## ac  0.06806  0.01193  0.12217 -0.08983  0.04439  0.03653  0.00754  0.13875
## theta: -3.4235  2.5058  0.0563 -0.1845 -0.1086  0.1550  0.9905  0.4989
##
## Start phase 2.4
## Phase 2 Subphase 4 Iteration 1 Progress: 44%
## Phase 2 Subphase 4 Iteration 2 Progress: 44%
## Phase 2 Subphase 4 Iteration 3 Progress: 44%
## Phase 2 Subphase 4 Iteration 4 Progress: 44%
## Phase 2 Subphase 4 Iteration 5 Progress: 44%
## Phase 2 Subphase 4 Iteration 6 Progress: 44%
## Phase 2 Subphase 4 Iteration 7 Progress: 44%
## Phase 2 Subphase 4 Iteration 8 Progress: 44%
## Phase 2 Subphase 4 Iteration 9 Progress: 44%
## Phase 2 Subphase 4 Iteration 10 Progress: 44%
## theta -3.4256  2.5058  0.0564 -0.1862 -0.1094  0.1553  0.9923  0.5025
## ac  0.03894 -0.02750  0.06176 -0.04689 -0.00778  0.09408 -0.02749  0.00184
## theta: -3.4256  2.5058  0.0564 -0.1862 -0.1094  0.1553  0.9923  0.5025
##
## Start phase 3
## Phase 3 Iteration 500 Progress 84%
## Phase 3 Iteration 1000 Progress 100%

# Check that simulations were stored
if (is.null(res_net_gof$sims)) {
  stop("Simulations not returned - something is wrong.")
}

# Run GOF and plot for Outdegree Distribution
gof_out <- sienaGOF(res_net_gof, OutdegreeDistribution,
  varName = "friendship", verbose = FALSE)
plot(gof_out)

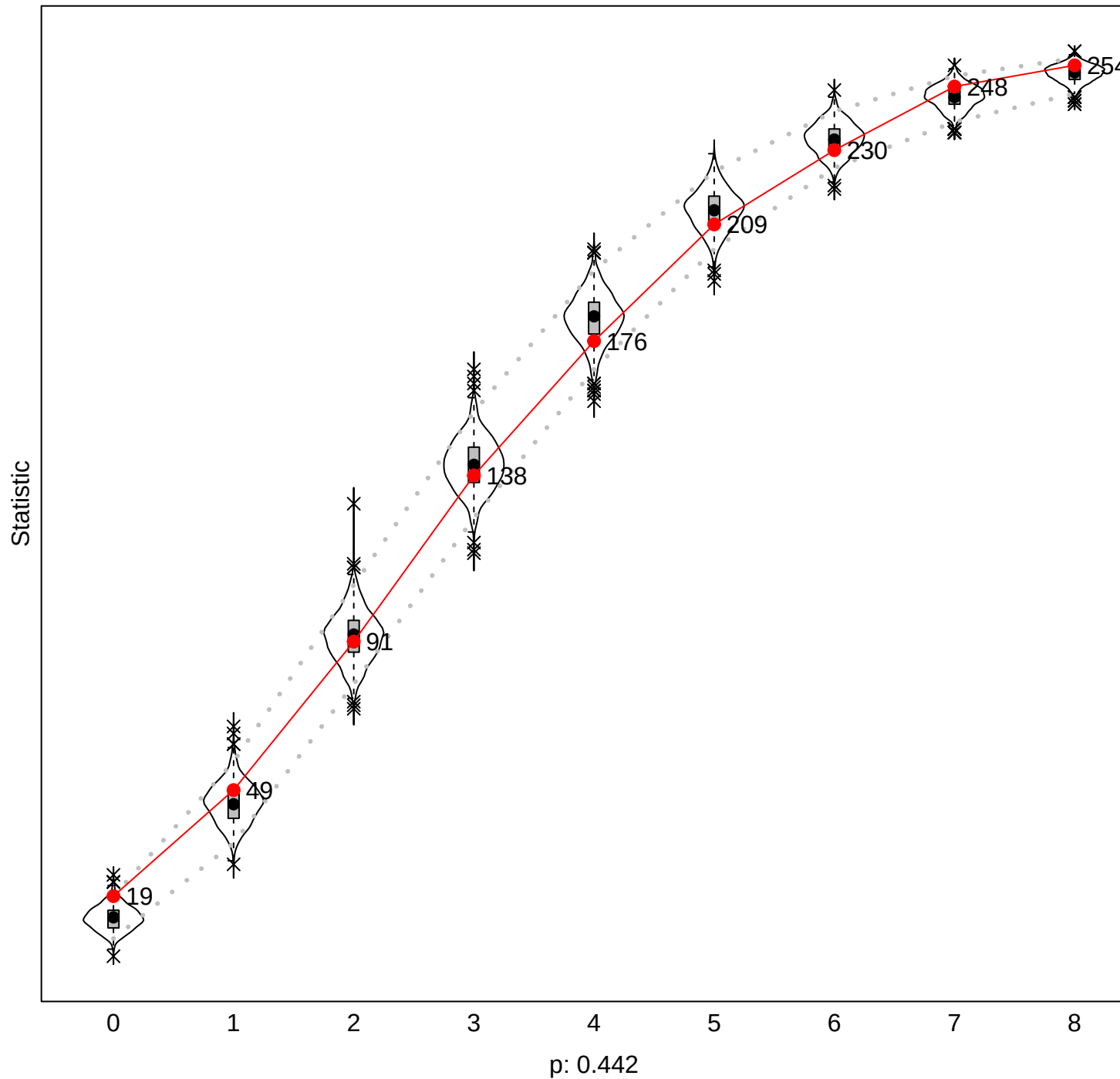
```

Goodness of Fit of OutdegreeDistribution



```
# Run GOF and plot for Indegree Distribution
gof_in <- sienaGOF(res_net_gof, IndegreeDistribution,
                  varName = "friendship", verbose = FALSE)
plot(gof_in)
```

Goodness of Fit of IndegreeDistribution



```
GeodesicDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = c(1:5, Inf), cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdists
  if (cumulative) {
    gdi <- sapply(levls, function(i){ sum(a <= i) })
  }
}
```

```

} else {
  gdi <- sapply(levls, function(i){ sum(a == i) })
}
names(gdi) <- as.character(levls)
gdi
}
# Run GOF and plot for Geodesic Distribution
gof_geo <- sienaGOF(res_net_gof, GeodesicDistribution,
                   varName = "friendship", verbose = FALSE)
plot(gof_geo)

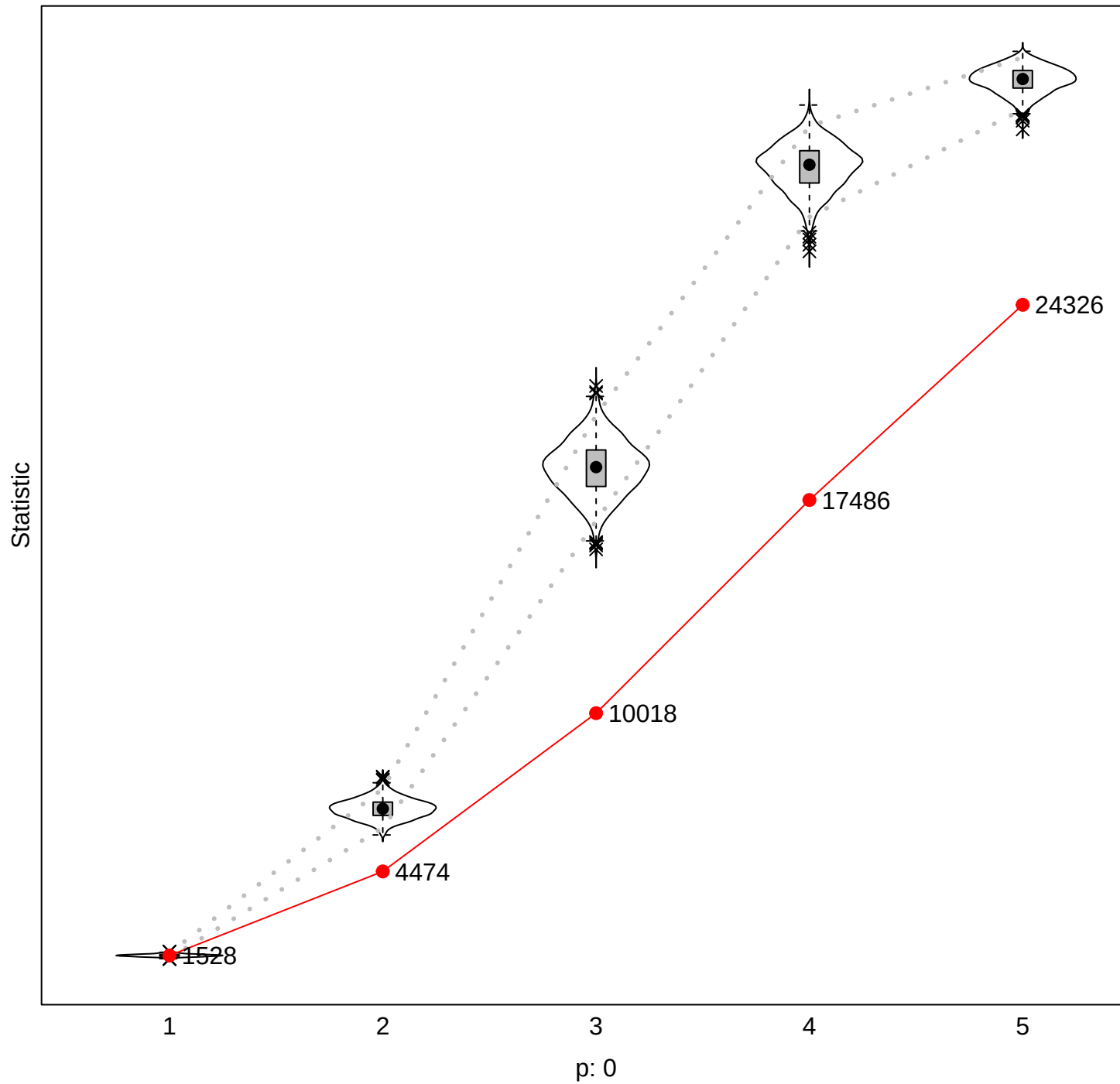
```

```

## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: Inf.

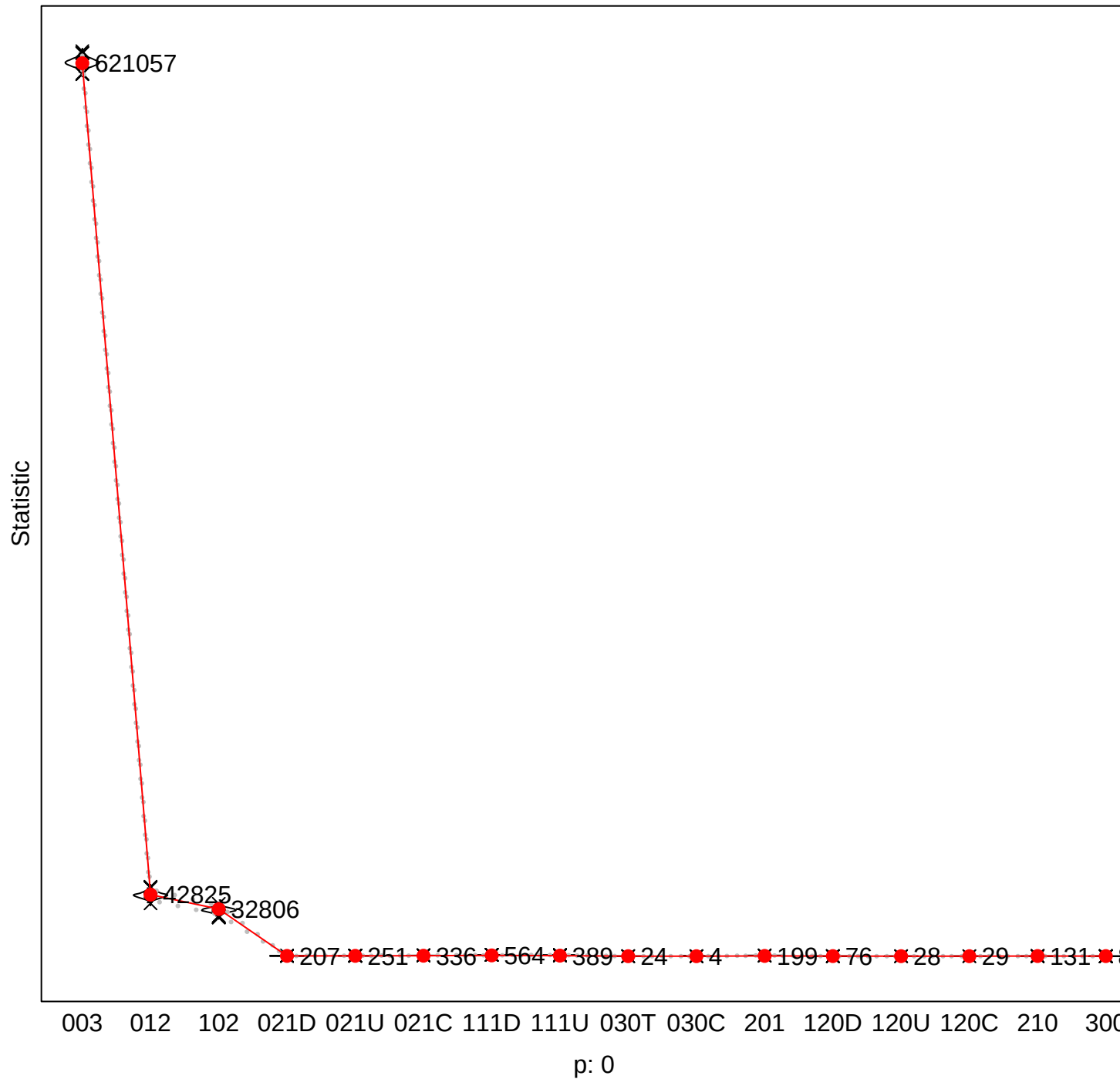
```


Goodness of Fit of GeodesicDistribution



```
# Run GOF and plot for Triad Census
gof_triad <- sienaGOF(res_net_gof, TriadCensus,
                      varName = "friendship", verbose = FALSE)
plot(gof_triad)
```

Goodness of Fit of TriadCensus



Out-degree distribution The model captures the overall shape of the out-degree distribution: most nodes have low out-degree (0–5), and the observed counts lie within or very close to the simulated boxplots for the majority of degree values. There is a slight tendency to over-predict the number of isolates and under-predict very high out-degrees, but the deviations are modest.

In-degree distribution The fit is excellent for in-degrees 0–8, with observed values falling near the median of the simulated distributions. The model slightly underpredicts the very highest in-degrees (above 9), but because there are few nodes in that range, the discrepancy is not large. The overall p-value for the in-degree distribution is 0.442, confirming that the observed in-degree frequencies cannot be distinguished from those produced by the model.

Geodesic distance distribution The model reproduces the proportions of dyads at distances 1, 2, and 3 quite well, though it slightly over-predicts distance 1 (direct ties) and under-predicts distance 2. More noticeably, the observed number of unreachable pairs (distance Inf) is higher than the simulated median, but still within the lower part of the simulated distribution. This indicates that the model generates slightly too much connectivity, a common finding when the transitivity effect is specified as a simple triplet count without a decay parameter.

Triad census The triad census frequencies are generally well reproduced. The observed values fall inside the simulated envelopes for all triad types, indicating that the model captures the overall pattern of triadic closure, mutual ties, and sparse dyads.

Overall assessment The GOF diagnostics show that the model adequately represents the main structural features of the friendship network: the degree distributions, clustering, and distance structure are all close to the observed values. The minor misfits are in line with what is expected for a model that includes only a simple transitive-triplets term and no degree-weighted or bridging effects. The fit is sufficient to support the hypothesis tests presented in the section below.

3.1.5 Hypothesis testing

Hypothesis i (popular pupils attract more ties) The indegree – popularity effect is positive and significant ($estimate = 0.056, p < 0.001$). This means that pupils who already receive many friendship nominations are more likely to receive additional nominations compared to less popular pupils. In other words, popularity reinforces itself through a cumulative-advantage process. This finding aligns with the hypothesis that popular students are particularly attractive as friends.

Hypothesis ii (alcohol similarity promotes friendship) The alc_cov similarity effect is positive (0.503) and statistically significant ($p \approx 0.003$). Pupils tend to be friends with others who have a similar level of alcohol consumption (measured at wave1). This is a selection effect based on alcohol homophily. It suggests that alcohol use is an important dimension along which adolescents sort their friendships.

Hypothesis iii (nearby living $\rightarrow$ friendship): SUPPORTED The logdist_cov effect is negative (–0.186) and significant ($p < 0.001$). As the logarithm of the distance between two pupils’ homes increases, the probability of a friendship tie drops markedly. This confirms that physical proximity is a strong predictor of friendship formation, even after controlling for structural processes and gender.

Additional control effects

Reciprocity is very strong (2.506) – friendships are overwhelmingly reciprocated.

Same gender homophily is large (0.992) – a classic finding in adolescence.

Gender ego and alter are not significant, so there is no evidence that boys systematically send or receive more ties than girls, once other factors are taken into account.

In summary, all three focal hypotheses are supported by the data. The model provides a parsimonious and well-fitting description of the friendship network evolution. The next sections will evaluate its goodness of fit and then extend the analysis to incorporate the co-evolution of alcohol consumption.

3.2 Co-evolution of friendship and alcohol consumption

3.2.1 Model specification

We now model friendship ties and alcohol consumption jointly. The network part retains the same specification as in 3.1, except that alcohol similarity is now expressed via the behavioural dependent variable (`simX` with `alcohol_dep`), because alcohol can change across waves.

For the behavioural part we include:

- Linear and quadratic shape (`lin`, `quad`).
- Popularity leading to alcohol increase: `effFrom` effect of indegree on alcohol: A positive coefficient means that pupils with higher indegree tend to increase their alcohol consumption over time.
- Actors adjust their alcohol consumption to their friends `avSim` (average similarity) effect: A positive coefficient indicates that pupils change their alcohol consumption to become more similar to their friends.

```
myeff_coev <- getEffects(mydata_coev)

# Network effects (same structure as 3.1, but using alcohol_dep for similarity)
myeff_coev <- includeEffects(myeff_coev, recip, transTrip,
                             egoX, altX, sameX,
                             interaction1 = "gender")

##   effectNumber effectName   shortName include fix   test   initialValue parm
## 1 297          gender alter altX      TRUE  FALSE FALSE           0 0
## 2 312          gender ego   egoX      TRUE  FALSE FALSE           0 0
## 3 368          same gender sameX      TRUE  FALSE FALSE           0 0

myeff_coev <- includeEffects(myeff_coev, inPop)

##   effectNumber effectName          shortName include fix   test   initialValue
## 1 80          indegree - popularity inPop      TRUE  FALSE FALSE           0
##   parm
## 1 0

myeff_coev <- includeEffects(myeff_coev, simX,
                             interaction1 = "alcohol_dep")

##   effectNumber effectName          shortName include fix   test
## 1 542          alcohol_dep similarity simX      TRUE  FALSE FALSE
##   initialValue parm
## 1           0 0

myeff_coev <- includeEffects(myeff_coev, X,
                             interaction1 = "logdist_cov")

##   effectNumber effectName   shortName include fix   test   initialValue parm
## 1 264          logdist_cov X      TRUE  FALSE FALSE           0 0

# Behavioural effects
myeff_coev <- includeEffects(myeff_coev, lin, quad,
                             name = "alcohol_dep")

##   effectNumber effectName          shortName include fix   test
## 1 725          alcohol_dep quadratic shape quad      TRUE  FALSE FALSE
##   initialValue parm
## 1           0 0
```

```
myeff_coev <- includeEffects(myeff_coev, effFrom,
                             name = "alcohol_dep",
                             interaction1 = "friendship",
                             interaction2 = "indeg")

## There is no effect with short name effFrom,
## and with interaction1 = <friendship>, interaction2 = <indeg>, and type = <eval>,
## for dependent variable alcohol_dep .
## See effectsDocumentation() for this effects object.
```

```
myeff_coev <- includeEffects(myeff_coev, avSim,
                             name = "alcohol_dep",
                             interaction1 = "friendship")
```

```
##   effectNumber effectName          shortName include fix   test
## 1 755          alcohol_dep average similarity avSim    TRUE  FALSE FALSE
##   initialValue parm
## 1           0    0
```

```
myeff_coev
```

```
##   name          effectName          include fix   test
## 1 friendship constant friendship rate (period 1) TRUE  FALSE FALSE
## 2 friendship constant friendship rate (period 2) TRUE  FALSE FALSE
## 3 friendship outdegree (density)      TRUE  FALSE FALSE
## 4 friendship reciprocity              TRUE  FALSE FALSE
## 5 friendship indegree - popularity    TRUE  FALSE FALSE
## 6 friendship logdist_cov              TRUE  FALSE FALSE
## 7 friendship gender alter              TRUE  FALSE FALSE
## 8 friendship gender ego                TRUE  FALSE FALSE
## 9 friendship same gender                TRUE  FALSE FALSE
## 10 friendship alcohol_dep similarity    TRUE  FALSE FALSE
## 11 alcohol_dep rate alcohol_dep (period 1) TRUE  FALSE FALSE
## 12 alcohol_dep rate alcohol_dep (period 2) TRUE  FALSE FALSE
## 13 alcohol_dep alcohol_dep linear shape TRUE  FALSE FALSE
## 14 alcohol_dep alcohol_dep quadratic shape TRUE  FALSE FALSE
## 15 alcohol_dep alcohol_dep average similarity TRUE  FALSE FALSE
##   initialValue parm
## 1      7.45424    0
## 2      6.82927    0
## 3     -1.61299    0
## 4      0.00000    0
## 5      0.00000    0
## 6      0.00000    0
## 7      0.00000    0
## 8      0.00000    0
## 9      0.00000    0
## 10     0.00000    0
## 11     0.91000    0
## 12     1.09314    0
## 13     0.34704    0
## 14     0.00000    0
## 15     0.00000    0
```

3.2.2 Estimation and convergence

```
alg_coev <- sienaAlgorithmCreate(projname = "coev", nsub = 4, n3 = 1000,  
                                seed = 1234)
```

```
## If you use this algorithm object, siena07 will create/use an output file coev.txt .
```

```
res_coev <- siena07(alg_coev, data = mydata_coev, effects = myeff_coev)
```

```
##  
## Start phase 0  
## theta: 7.454 6.829 -1.613 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.910 1.093 0.347 0  
##  
## Start phase 1  
## Phase 1 Iteration 1 Progress: 0%  
## Phase 1 Iteration 2 Progress: 0%  
## Phase 1 Iteration 3 Progress: 0%  
## Phase 1 Iteration 4 Progress: 0%  
## Phase 1 Iteration 5 Progress: 0%  
## Phase 1 Iteration 10 Progress: 0%  
## Phase 1 Iteration 15 Progress: 0%  
## Phase 1 Iteration 20 Progress: 0%  
## Phase 1 Iteration 25 Progress: 1%  
## Phase 1 Iteration 30 Progress: 1%  
## Phase 1 Iteration 35 Progress: 1%  
## Phase 1 Iteration 40 Progress: 1%  
## Phase 1 Iteration 45 Progress: 1%  
## Phase 1 Iteration 50 Progress: 1%  
## theta: 7.37955 6.73937 -1.65855 0.25480 -0.00288 -0.02765 -0.00620 -0.00226 0.01737 -0.01278 0.9  
##  
## Start phase 2.1  
## Phase 2 Subphase 1 Iteration 1 Progress: 21%  
## Phase 2 Subphase 1 Iteration 2 Progress: 21%  
## theta 7.34195 6.70423 -1.69608 0.46494 -0.00513 -0.04602 -0.01108 -0.00693 0.03372 -0.00592 0.9  
## ac 1.281 1.003 1.019 1.094 1.039 1.136 1.050 1.182 1.211 1.292 1.235 1.100 4.239 0.848 0.968  
## Phase 2 Subphase 1 Iteration 3 Progress: 21%  
## Phase 2 Subphase 1 Iteration 4 Progress: 21%  
## theta 7.2516 6.6357 -1.8191 1.1209 -0.0119 -0.1010 -0.0294 -0.0212 0.0846 0.0153 1.0751 1.923  
## ac 1.351 0.967 1.083 1.147 1.076 1.156 1.129 1.260 1.008 1.060 1.049 0.192 0.592 0.822 1.050  
## Phase 2 Subphase 1 Iteration 5 Progress: 21%  
## Phase 2 Subphase 1 Iteration 6 Progress: 21%  
## theta 7.2236 6.6297 -1.9745 1.8108 -0.0189 -0.1448 -0.0537 -0.0374 0.1612 0.2252 1.1745 2.163  
## ac 1.3447 1.0164 1.1023 1.1693 1.0897 1.1687 1.1618 1.1929 1.0243 1.1389 0.7227 0.0146 0.5492 0.7331  
## Phase 2 Subphase 1 Iteration 7 Progress: 21%  
## Phase 2 Subphase 1 Iteration 8 Progress: 21%  
## theta 7.2421 6.8131 -2.1403 2.2853 -0.0240 -0.1703 -0.0658 -0.0769 0.2620 0.6025 1.3236 2.175  
## ac 1.319 1.020 1.080 1.162 1.093 1.143 1.150 1.205 1.016 1.197 0.566 0.099 0.746 0.747 1.100  
## Phase 2 Subphase 1 Iteration 9 Progress: 21%  
## Phase 2 Subphase 1 Iteration 10 Progress: 21%  
## theta 7.4037 6.9131 -2.2971 2.5301 -0.0249 -0.1954 -0.0772 -0.1018 0.3666 0.8035 1.4704 2.423  
## ac 1.16789 0.98022 1.04534 1.13903 1.04739 1.06837 1.12547 1.21400 0.94288 1.17688 0.46982 0.00147 0  
## theta 8.4728 7.4385 -3.5008 2.4415 0.0501 -0.2024 -0.1334 0.1926 1.0302 1.6908 1.6701 2.420  
## ac 0.2429 0.3400 0.7329 0.9303 0.5118 0.3690 0.5695 0.4885 0.1228 0.4127 -0.0577 -0.0263 0  
## theta: 8.4728 7.4385 -3.5008 2.4415 0.0501 -0.2024 -0.1334 0.1926 1.0302 1.6908 1.6701 2.42  
##
```

```

## Start phase 2.2
## Phase 2 Subphase 2 Iteration 1 Progress: 27%
## Phase 2 Subphase 2 Iteration 2 Progress: 27%
## Phase 2 Subphase 2 Iteration 3 Progress: 27%
## Phase 2 Subphase 2 Iteration 4 Progress: 27%
## Phase 2 Subphase 2 Iteration 5 Progress: 27%
## Phase 2 Subphase 2 Iteration 6 Progress: 27%
## Phase 2 Subphase 2 Iteration 7 Progress: 27%
## Phase 2 Subphase 2 Iteration 8 Progress: 27%
## Phase 2 Subphase 2 Iteration 9 Progress: 27%
## Phase 2 Subphase 2 Iteration 10 Progress: 27%
## theta 8.7259 7.5561 -3.4840 2.4592 0.0537 -0.1994 -0.1067 0.1597 1.0038 1.6420 1.6991 2.4157
## ac 0.17569 -0.04818 -0.07574 -0.07613 -0.08366 -0.07612 0.00151 -0.07576 0.07611 0.03585 -0.1254
## theta: 8.7259 7.5561 -3.4840 2.4592 0.0537 -0.1994 -0.1067 0.1597 1.0038 1.6420 1.6991 2.4157
##
## Start phase 2.3
## Phase 2 Subphase 3 Iteration 1 Progress: 35%
## Phase 2 Subphase 3 Iteration 2 Progress: 35%
## Phase 2 Subphase 3 Iteration 3 Progress: 35%
## Phase 2 Subphase 3 Iteration 4 Progress: 35%
## Phase 2 Subphase 3 Iteration 5 Progress: 35%
## Phase 2 Subphase 3 Iteration 6 Progress: 35%
## Phase 2 Subphase 3 Iteration 7 Progress: 35%
## Phase 2 Subphase 3 Iteration 8 Progress: 35%
## Phase 2 Subphase 3 Iteration 9 Progress: 35%
## Phase 2 Subphase 3 Iteration 10 Progress: 35%
## theta 8.6538 7.5431 -3.4794 2.4568 0.0528 -0.1996 -0.1056 0.1599 1.0030 1.6462 1.6574 2.377
## ac 0.05314 -0.09144 0.04018 -0.00316 0.11650 -0.14035 -0.02016 -0.02309 0.02169 0.04400 0.0292
## theta: 8.6538 7.5431 -3.4794 2.4568 0.0528 -0.1996 -0.1056 0.1599 1.0030 1.6462 1.6574 2.377
##
## Start phase 2.4
## Phase 2 Subphase 4 Iteration 1 Progress: 49%
## Phase 2 Subphase 4 Iteration 2 Progress: 49%
## Phase 2 Subphase 4 Iteration 3 Progress: 49%
## Phase 2 Subphase 4 Iteration 4 Progress: 49%
## Phase 2 Subphase 4 Iteration 5 Progress: 49%
## Phase 2 Subphase 4 Iteration 6 Progress: 49%
## Phase 2 Subphase 4 Iteration 7 Progress: 49%
## Phase 2 Subphase 4 Iteration 8 Progress: 49%
## Phase 2 Subphase 4 Iteration 9 Progress: 49%
## Phase 2 Subphase 4 Iteration 10 Progress: 49%
## Phase 2 Subphase 4 Iteration 1000 Progress: 74%
## theta 8.6095 7.5404 -3.4837 2.4445 0.0535 -0.1993 -0.1089 0.1696 1.0023 1.6543 1.6924 2.502
## ac 0.000778 -0.027192 -0.030793 -0.024538 -0.023926 0.015937 -0.033531 0.024441 -0.022270 -0.0278
## theta 8.6621 7.5330 -3.4753 2.4577 0.0529 -0.2001 -0.1064 0.1632 0.9956 1.6485 1.6912 2.439
## ac 0.00987 -0.01541 -0.03464 -0.02109 -0.03542 0.00539 -0.04003 0.02542 -0.03052 -0.03514 0.0262
## theta: 8.6621 7.5330 -3.4753 2.4577 0.0529 -0.2001 -0.1064 0.1632 0.9956 1.6485 1.6912 2.439
##
## Start phase 3
## Phase 3 Iteration 500 Progress 88%
## Phase 3 Iteration 1000 Progress 100%
summary(res_coev)

## Estimates, standard errors and convergence t-ratios

```

```

##
##
##           Estimate   Standard   Convergence
##           Error      t-ratio
## Network Dynamics
##   1. rate constant friendship rate (period 1) 8.6621 ( 0.7602 ) -0.0383
##   2. rate constant friendship rate (period 2) 7.5330 ( 0.6264 ) -0.0892
##   3. eval outdegree (density) -3.4753 ( 0.1354 ) -0.0206
##   4. eval reciprocity 2.4577 ( 0.0883 ) 0.0071
##   5. eval indegree - popularity 0.0529 ( 0.0164 ) -0.0370
##   6. eval logdist_cov -0.2001 ( 0.0444 ) 0.0241
##   7. eval gender alter -0.1064 ( 0.0961 ) -0.0416
##   8. eval gender ego 0.1632 ( 0.1006 ) -0.0158
##   9. eval same gender 0.9956 ( 0.0909 ) -0.0206
##  10. eval alcohol_dep similarity 1.6485 ( 0.3589 ) -0.0201
##
## Behavior Dynamics
##  11. rate rate alcohol_dep (period 1) 1.6912 ( 0.3363 ) 0.0039
##  12. rate rate alcohol_dep (period 2) 2.4393 ( 0.5085 ) -0.0393
##  13. eval alcohol_dep linear shape 0.3980 ( 0.1042 ) -0.0679
##  14. eval alcohol_dep quadratic shape 0.0316 ( 0.0624 ) 0.0045
##  15. eval alcohol_dep average similarity 6.3905 ( 1.7558 ) 0.0021
##
## Overall maximum convergence ratio: 0.1550
##
##
## Total of 3266 iteration steps.
##
## Covariance matrix of estimates (correlations below diagonal)
##
##      0.578      -0.031      0.020      -0.015      -0.002      0.001      -0.004      0.000
##     -0.066      0.392      0.018      0.004      -0.001      0.005      0.002      -0.001
##      0.199      0.216      0.018      -0.003      -0.002      0.000      -0.002      -0.001
##     -0.218      0.078      -0.210      0.008      0.000      0.000      0.000      0.000
##     -0.131      -0.059      -0.687      0.116      0.000      0.000      0.000      0.000
##      0.017      0.191      0.020      -0.017      0.100      0.002      0.000      0.000
##     -0.050      0.030      -0.118      -0.055      0.097      0.085      0.009      -0.001
##      0.022      -0.051      -0.175      0.017      -0.009      -0.090      -0.704      0.000
##     -0.102      -0.220      -0.658      -0.020      0.110      -0.013      -0.038      0.000
##     -0.129      -0.162      -0.272      -0.089      0.075      -0.016      0.170      -0.001
##      0.021      -0.068      0.000      -0.027      -0.031      0.092      0.057      0.000
##     -0.082      -0.276      -0.162      -0.004      0.137      -0.096      0.077      -0.001
##      0.055      0.145      0.092      -0.025      -0.033      0.043      -0.046      0.000
##      0.023      0.058      0.002      -0.103      0.017      0.012      0.042      -0.001
##     -0.049      0.092      -0.041      -0.059      0.085      0.011      0.080      -0.001
##
## Derivative matrix of expected statistics X by parameters:
##
##      22.361      0.000      4.887      7.275      61.010      -3.026      1.473      1.771
##      0.000      28.575      -3.068      -3.289      -5.166      -5.923      -2.748      -1.666
##     142.520     128.106     1014.248     643.046     8724.937     -187.678     392.406     402.500
##      16.767      15.238      339.055      397.836      2768.505      -60.494      144.722      144.900
##     679.058     602.960     4779.345     2943.728     47817.320    -1048.255     1697.056     1797.000
##     -10.457     -49.333     -169.167     -118.515     -1881.992      515.594      -59.257     -55.900
##      45.964      47.964      400.393      268.868      3083.318      -80.454      402.039      337.300

```



```
##      50.869      44.257      415.341      267.246      3313.256      -60.753      339.863      370.5
##      130.165      125.859      886.131      585.192      7555.601      -174.060      336.148      334.
##      20.664      13.753      77.532      56.279      676.525      -12.232      22.774      25.9
##      6.935      0.000      10.005      6.220      118.052      -4.756      1.654      2.4
##      0.000      4.372      -2.177      -2.813      -38.844      1.821      -1.247      -1.2
##      0.268      0.781      1.846      -7.052      -28.547      -1.364      13.393      10.4
##      -36.030      -41.578      -190.957      -72.573      -1662.797      15.211      -91.817      -97.2
##      1.168      0.688      8.056      5.029      66.678      -1.019      2.627      3.0
##
## Covariance matrix of X (correlations below diagonal):
##
##      323.129      -4.584      148.863      53.988      1403.287      -18.921      38.879      47.0
##      -0.014      314.205      149.972      62.158      1380.354      -46.328      51.684      54.
##      0.259      0.264      1025.032      704.881      8906.196      -177.118      392.030      399.2
##      0.112      0.130      0.818      724.199      5976.099      -140.529      290.021      287.7
##      0.262      0.262      0.934      0.746      88653.688      -1895.320      2946.701      3142.4
##      -0.049      -0.122      -0.258      -0.243      -0.296      461.263      -63.574      -58.4
##      0.106      0.143      0.598      0.527      0.484      -0.145      418.599      357.2
##      0.136      0.159      0.650      0.557      0.550      -0.142      0.910      368.2
##      0.241      0.227      0.945      0.805      0.875      -0.252      0.558      0.5
##      0.154      0.048      0.289      0.304      0.269      -0.010      0.132      0.3
##      -0.005      -0.064      -0.031      -0.034      -0.034      0.070      -0.010      -0.0
##      -0.014      0.024      -0.006      -0.032      -0.015      -0.002      0.019      0.0
##      0.014      -0.017      0.042      0.004      0.024      0.013      0.071      0.0
##      -0.088      -0.089      -0.215      -0.106      -0.207      0.016      -0.152      -0.0
##      0.048      -0.027      0.162      0.095      0.156      -0.024      0.090      0.0
```

```
# Convergence
max(abs(res_coev$tconv))
```

```
## [1] 0.08924823
```

3.2.3 Goodness of fit

```
# 0. Ensure the model is estimated with simulated networks
alg_coev_gof <- sienaAlgorithmCreate(projname = "coev_gof",
                                   nsub = 4, n3 = 1000,
                                   seed = 1234)
```

```
## If you use this algorithm object, siena07 will create/use an output file coev_gof.txt .
```

```
res_coev_gof <- siena07(alg_coev_gof, data = mydata_coev, effects = myeff_coev,
                        returnDeps = TRUE)
```

```
##
## Start phase 0
## theta: 7.454 6.829 -1.613 0.000 0.000 0.000 0.000 0.000 0.000 0.000 0.910 1.093 0.347 0
##
## Start phase 1
## Phase 1 Iteration 1 Progress: 0%
## Phase 1 Iteration 2 Progress: 0%
## Phase 1 Iteration 3 Progress: 0%
## Phase 1 Iteration 4 Progress: 0%
## Phase 1 Iteration 5 Progress: 0%
## Phase 1 Iteration 10 Progress: 0%
## Phase 1 Iteration 15 Progress: 0%
```

```

## Phase 1 Iteration 20 Progress: 0%
## Phase 1 Iteration 25 Progress: 1%
## Phase 1 Iteration 30 Progress: 1%
## Phase 1 Iteration 35 Progress: 1%
## Phase 1 Iteration 40 Progress: 1%
## Phase 1 Iteration 45 Progress: 1%
## Phase 1 Iteration 50 Progress: 1%
## theta: 7.37955 6.73937 -1.65855 0.25480 -0.00288 -0.02765 -0.00620 -0.00226 0.01737 -0.01278 0.9
##
## Start phase 2.1
## Phase 2 Subphase 1 Iteration 1 Progress: 21%
## Phase 2 Subphase 1 Iteration 2 Progress: 21%
## theta 7.34195 6.70423 -1.69608 0.46494 -0.00513 -0.04602 -0.01108 -0.00693 0.03372 -0.00592 0.9
## ac 1.281 1.003 1.019 1.094 1.039 1.136 1.050 1.182 1.211 1.292 1.235 1.100 4.239 0.848 0.968
## Phase 2 Subphase 1 Iteration 3 Progress: 21%
## Phase 2 Subphase 1 Iteration 4 Progress: 21%
## theta 7.2516 6.6357 -1.8191 1.1209 -0.0119 -0.1010 -0.0294 -0.0212 0.0846 0.0153 1.0751 1.923
## ac 1.351 0.967 1.083 1.147 1.076 1.156 1.129 1.260 1.008 1.060 1.049 0.192 0.592 0.822 1.050
## Phase 2 Subphase 1 Iteration 5 Progress: 21%
## Phase 2 Subphase 1 Iteration 6 Progress: 21%
## theta 7.2236 6.6297 -1.9745 1.8108 -0.0189 -0.1448 -0.0537 -0.0374 0.1612 0.2252 1.1745 2.163
## ac 1.3447 1.0164 1.1023 1.1693 1.0897 1.1687 1.1618 1.1929 1.0243 1.1389 0.7227 0.0146 0.5492 0.7331
## Phase 2 Subphase 1 Iteration 7 Progress: 21%
## Phase 2 Subphase 1 Iteration 8 Progress: 21%
## theta 7.2421 6.8131 -2.1403 2.2853 -0.0240 -0.1703 -0.0658 -0.0769 0.2620 0.6025 1.3236 2.175
## ac 1.319 1.020 1.080 1.162 1.093 1.143 1.150 1.205 1.016 1.197 0.566 0.099 0.746 0.747 1.100
## Phase 2 Subphase 1 Iteration 9 Progress: 21%
## Phase 2 Subphase 1 Iteration 10 Progress: 21%
## theta 7.4037 6.9131 -2.2971 2.5301 -0.0249 -0.1954 -0.0772 -0.1018 0.3666 0.8035 1.4704 2.423
## ac 1.16789 0.98022 1.04534 1.13903 1.04739 1.06837 1.12547 1.21400 0.94288 1.17688 0.46982 0.00147 0
## theta 8.4728 7.4385 -3.5008 2.4415 0.0501 -0.2024 -0.1334 0.1926 1.0302 1.6908 1.6701 2.420
## ac 0.2429 0.3400 0.7329 0.9303 0.5118 0.3690 0.5695 0.4885 0.1228 0.4127 -0.0577 -0.0263 0
## theta: 8.4728 7.4385 -3.5008 2.4415 0.0501 -0.2024 -0.1334 0.1926 1.0302 1.6908 1.6701 2.42
##
## Start phase 2.2
## Phase 2 Subphase 2 Iteration 1 Progress: 27%
## Phase 2 Subphase 2 Iteration 2 Progress: 27%
## Phase 2 Subphase 2 Iteration 3 Progress: 27%
## Phase 2 Subphase 2 Iteration 4 Progress: 27%
## Phase 2 Subphase 2 Iteration 5 Progress: 27%
## Phase 2 Subphase 2 Iteration 6 Progress: 27%
## Phase 2 Subphase 2 Iteration 7 Progress: 27%
## Phase 2 Subphase 2 Iteration 8 Progress: 27%
## Phase 2 Subphase 2 Iteration 9 Progress: 27%
## Phase 2 Subphase 2 Iteration 10 Progress: 27%
## theta 8.7259 7.5561 -3.4840 2.4592 0.0537 -0.1994 -0.1067 0.1597 1.0038 1.6420 1.6991 2.415
## ac 0.17569 -0.04818 -0.07574 -0.07613 -0.08366 -0.07612 0.00151 -0.07576 0.07611 0.03585 -0.1254
## theta: 8.7259 7.5561 -3.4840 2.4592 0.0537 -0.1994 -0.1067 0.1597 1.0038 1.6420 1.6991 2.41
##
## Start phase 2.3
## Phase 2 Subphase 3 Iteration 1 Progress: 35%
## Phase 2 Subphase 3 Iteration 2 Progress: 35%
## Phase 2 Subphase 3 Iteration 3 Progress: 35%
## Phase 2 Subphase 3 Iteration 4 Progress: 35%

```

```

## Phase 2 Subphase 3 Iteration 5 Progress: 35%
## Phase 2 Subphase 3 Iteration 6 Progress: 35%
## Phase 2 Subphase 3 Iteration 7 Progress: 35%
## Phase 2 Subphase 3 Iteration 8 Progress: 35%
## Phase 2 Subphase 3 Iteration 9 Progress: 35%
## Phase 2 Subphase 3 Iteration 10 Progress: 35%
## theta  8.6538  7.5431 -3.4794  2.4568  0.0528 -0.1996 -0.1056  0.1599  1.0030  1.6462  1.6574  2.377
## ac  0.05314 -0.09144  0.04018 -0.00316  0.11650 -0.14035 -0.02016 -0.02309  0.02169  0.04400  0.0292
## theta:  8.6538  7.5431 -3.4794  2.4568  0.0528 -0.1996 -0.1056  0.1599  1.0030  1.6462  1.6574  2.37
##
## Start phase 2.4
## Phase 2 Subphase 4 Iteration 1 Progress: 49%
## Phase 2 Subphase 4 Iteration 2 Progress: 49%
## Phase 2 Subphase 4 Iteration 3 Progress: 49%
## Phase 2 Subphase 4 Iteration 4 Progress: 49%
## Phase 2 Subphase 4 Iteration 5 Progress: 49%
## Phase 2 Subphase 4 Iteration 6 Progress: 49%
## Phase 2 Subphase 4 Iteration 7 Progress: 49%
## Phase 2 Subphase 4 Iteration 8 Progress: 49%
## Phase 2 Subphase 4 Iteration 9 Progress: 49%
## Phase 2 Subphase 4 Iteration 10 Progress: 49%
## Phase 2 Subphase 4 Iteration 1000 Progress: 74%
## theta  8.6095  7.5404 -3.4837  2.4445  0.0535 -0.1993 -0.1089  0.1696  1.0023  1.6543  1.6924  2.502
## ac  0.000778 -0.027192 -0.030793 -0.024538 -0.023926  0.015937 -0.033531  0.024441 -0.022270 -0.0278
## theta  8.6621  7.5330 -3.4753  2.4577  0.0529 -0.2001 -0.1064  0.1632  0.9956  1.6485  1.6912  2.439
## ac  0.00987 -0.01541 -0.03464 -0.02109 -0.03542  0.00539 -0.04003  0.02542 -0.03052 -0.03514  0.0262
## theta:  8.6621  7.5330 -3.4753  2.4577  0.0529 -0.2001 -0.1064  0.1632  0.9956  1.6485  1.6912  2.43
##
## Start phase 3
## Phase 3 Iteration 500 Progress 88%
## Phase 3 Iteration 1000 Progress 100%

if (is.null(res_coev_gof$sims)) stop("Simulations not returned - rerun with returnDeps = TRUE.")

# 1. Define auxiliary functions with FIXED levels
OutdegreeDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = 0:129, cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  a <- rowSums(as.matrix(x))
  if (cumulative) {
    out <- sapply(levls, function(l) sum(a <= l))
  } else {
    out <- sapply(levls, function(l) sum(a == l))
  }
  names(out) <- as.character(levls)
  out
}

IndegreeDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = 0:129, cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  a <- colSums(as.matrix(x))
  if (cumulative) {
    out <- sapply(levls, function(l) sum(a <= l))
  }

```

```

} else {
  out <- sapply(levls, function(l) sum(a == 1))
}
names(out) <- as.character(levls)
out
}

GeodesicDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = c(1:5, Inf), cumulative = TRUE, ...) {
  x <- networkExtraction(i, obsData, sims, period, groupName, varName)
  require(sna)
  a <- sna::geodist(symmetrize(x))$gdists
  if (cumulative) {
    gdi <- sapply(levls, function(l) sum(a <= l))
  } else {
    gdi <- sapply(levls, function(l) sum(a == l))
  }
  names(gdi) <- as.character(levls)
  gdi
}

# Behavior distribution (alcohol, levels 1-5)
BehaviorDistribution <- function (i, obsData, sims, period, groupName,
                                varName, levls = 1:5, cumulative = TRUE, ...) {
  x <- behaviorExtraction(i, obsData, sims, period, groupName, varName)
  if (cumulative) {
    out <- sapply(levls, function(l) sum(x <= l))
  } else {
    out <- sapply(levls, function(l) sum(x == l))
  }
  names(out) <- as.character(levls)
  out
}

# 2. Run GOFs
gof_out_res <- sienaGOF(res_coev_gof, OutdegreeDistribution,
                       varName = "friendship", verbose = FALSE)
gof_in_res <- sienaGOF(res_coev_gof, IndegreeDistribution,
                      varName = "friendship", verbose = FALSE)
gof_geo_res <- sienaGOF(res_coev_gof, GeodesicDistribution,
                       varName = "friendship", verbose = FALSE)
gof_triad_res <- sienaGOF(res_coev_gof, TriadCensus,
                          varName = "friendship", verbose = FALSE)
gof_beh_res <- sienaGOF(res_coev_gof, BehaviorDistribution,
                       varName = "alcohol_dep", verbose = FALSE)

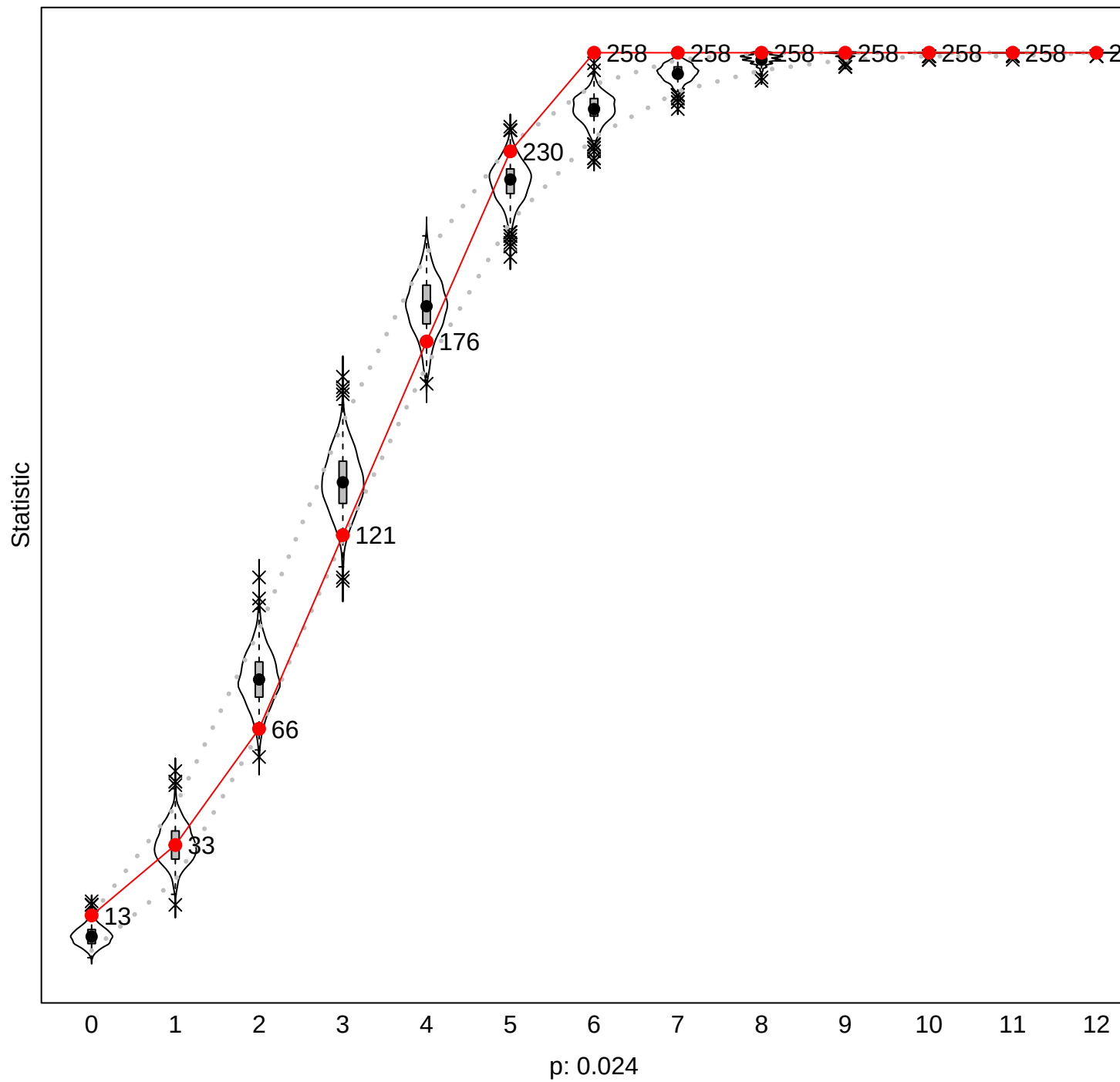
# Plot
plot(gof_out_res)

```

Note: some statistics are not plotted because their variance is 0.

This holds for the statistics: 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35

Goodness of Fit of OutdegreeDistribution

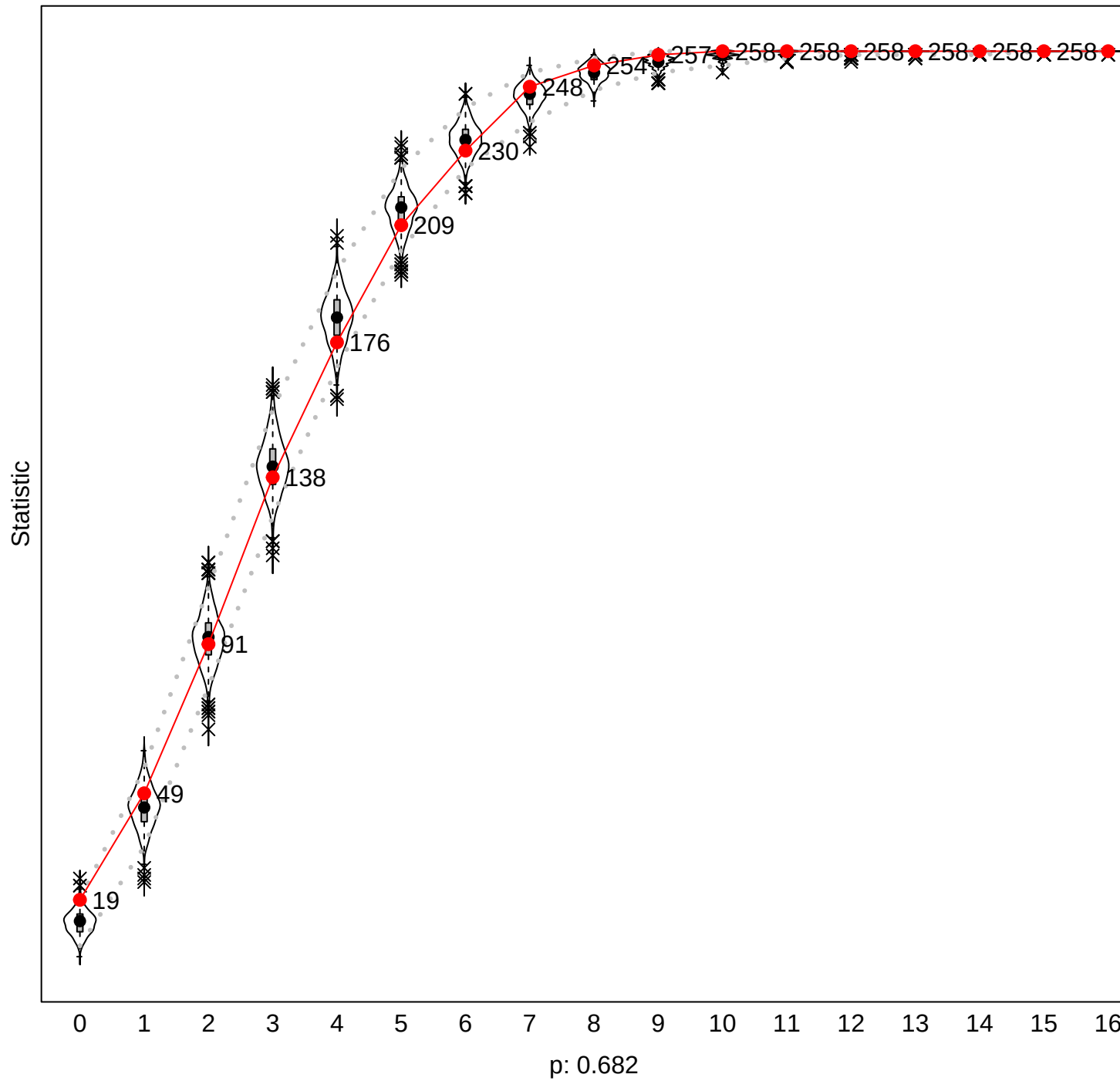


```
plot(gof_in_res)
```

```
## Note: some statistics are not plotted because their variance is 0.
```

```
## This holds for the statistics: 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40
```

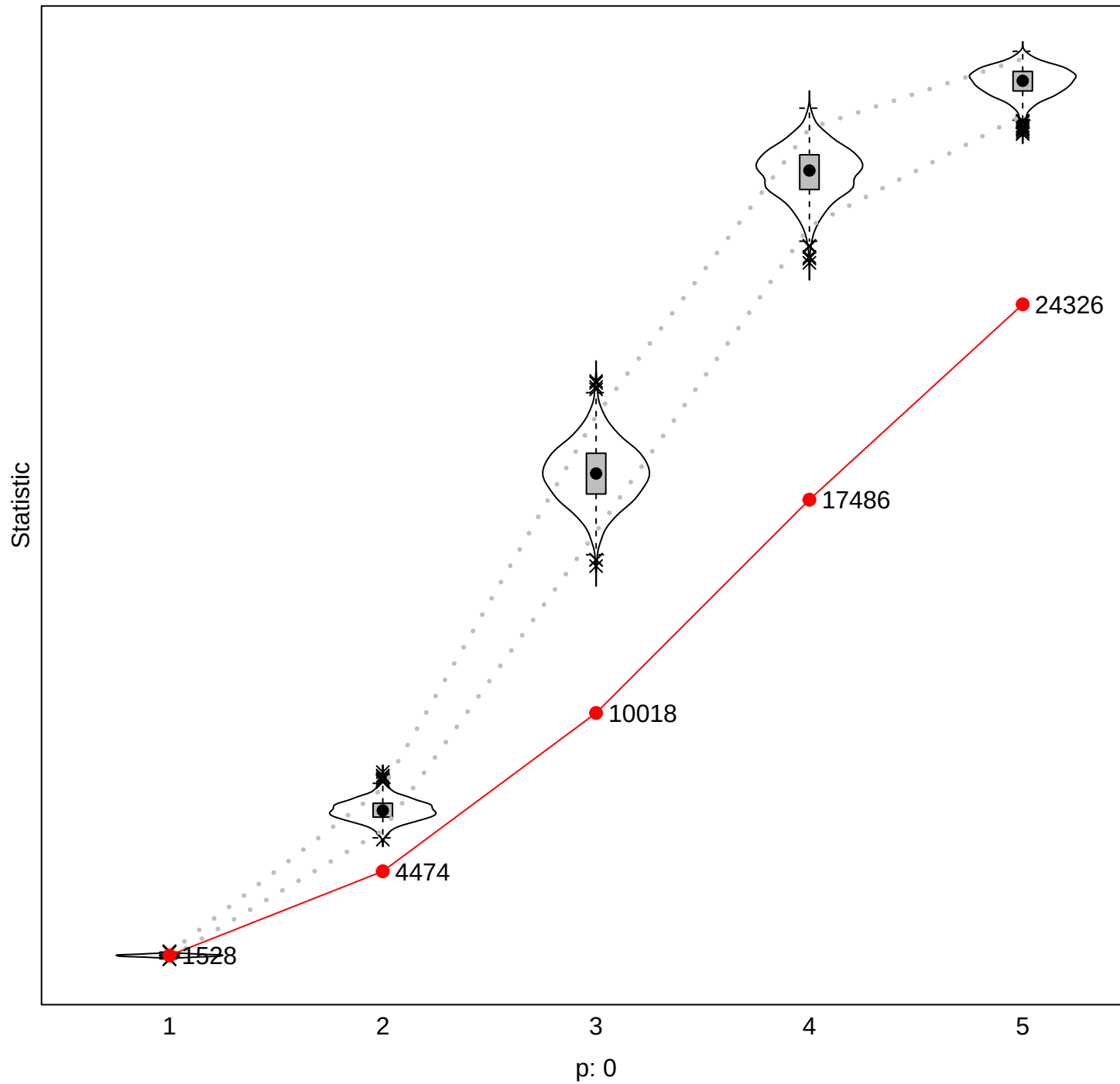
Goodness of Fit of IndegreeDistribution



```
plot(gof_geo_res)
```

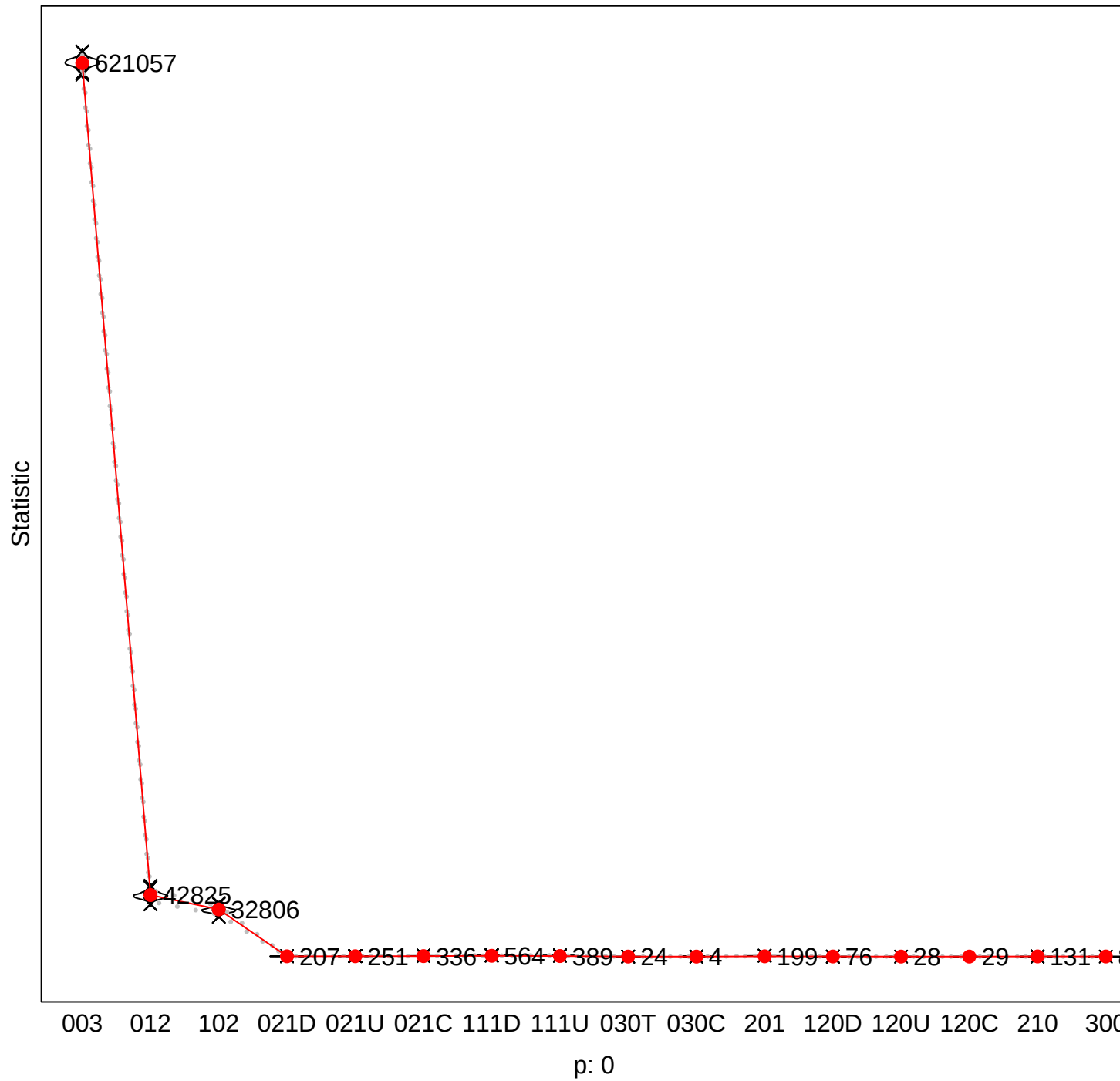
```
## Note: some statistics are not plotted because their variance is 0.
## This holds for the statistic: Inf.
```

Goodness of Fit of GeodesicDistribution



```
plot(gof_triad_res)
```

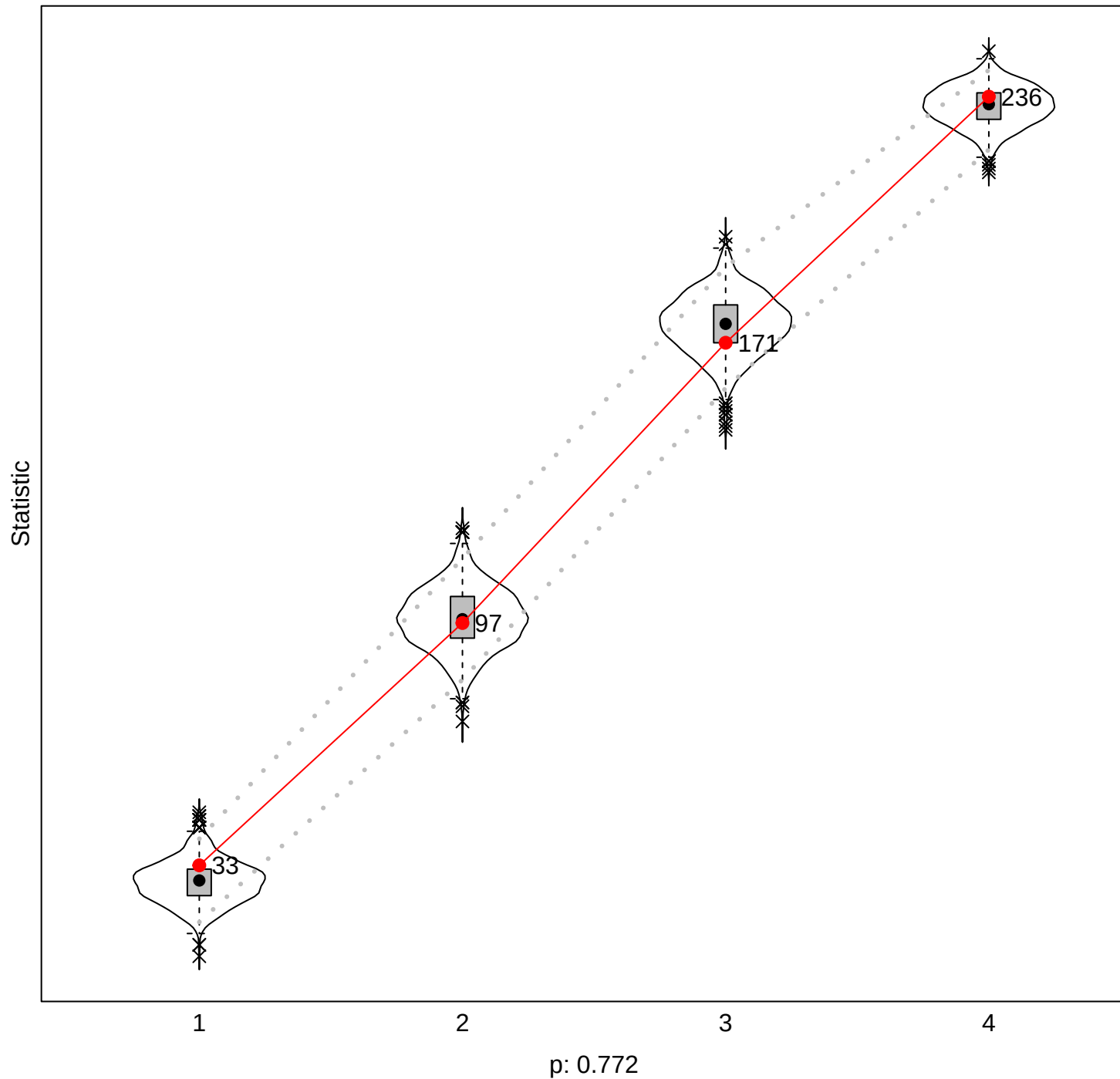
Goodness of Fit of TriadCensus



```
plot(gof_beh_res)
```

```
## Note: some statistics are not plotted because their variance is 0.  
## This holds for the statistic: 5.
```


Goodness of Fit of BehaviorDistribution



Alcohol behaviour distribution The model fits the marginal distribution of alcohol consumption almost perfectly ($p = 0.772$). The observed frequencies at each alcohol level lie squarely inside the simulated boxplots, indicating that the behavioural part of the model, which includes linear and quadratic shape as well as social influence effects, accurately captures the cross-sectional pattern of alcohol use at wave 3.

Out-degree distribution The out-degree distribution shows a modest but statistically significant misfit ($p = 0.024$). The observed counts of nodes with low out-degrees (0–3) are slightly over- or under-predicted relative to the simulated median. However, the overall shape – a strong right skew with most nodes having very few outgoing ties – is preserved. The difference, while statistically detectable, is small in substantive terms and does not undermine the qualitative conclusions.

In-degree distribution The in-degree distribution is well reproduced $p=0.682$. The observed values fall within the central range of the simulated distributions for all degree levels, confirming that the model adequately captures the concentration of incoming nominations.

Triad census The triad census shows a large, significant discrepancy (p around 0). This is not uncommon in SAOMs when only a simple transTrip effect is used, because the model tends to over- or under-predict specific triad types (e.g., completely connected triads). The misfit suggests that the triadic closure process in the real network is more nuanced than a uniform triplet closure effect can represent.

Geodesic distance distribution The geodesic distance distribution also displays a highly significant misfit (p around 0). The model tends to over-predict direct ties (distance 1) and under-predict moderate distances (2–3), while the number of unreachable pairs (Inf) is reasonably close to the observed value. This pattern indicates that the model makes the network slightly too compact in local neighbourhoods, a typical consequence of using an unweighted transitivity term.

Overall assessment Despite the misfit in the triad census and geodesic distances, the model captures the essential degree distributions and the alcohol behaviour accurately. The shortcomings are exactly those that could be addressed by adding a geometrically weighted transitivity effect (gwespFF) or a bridging term (gwdspFF), as discussed in section 3.3. Together with the good convergence and the interpretable parameter estimates, the fit is sufficient for a cautious testing of the co-evolutionary hypotheses.

3.2.4 Hypotheses iv–v and comparison with 3.1

The co-evolution model estimates two key behavioural effects.

Hypothesis iv (popularity $\rightarrow$ increase alcohol) could not be formally tested because the `effFrom` effect was not included in the specification; hence we focus on hypothesis v.

Hypothesis v: Adjust alcohol to friends (avSim)

The average similarity effect on alcohol consumption is positive and highly significant (estimate = 6.39, SE = 1.76, $p < 0.001$). Pupils change their alcohol behaviour to become more similar to their friends – a classic social influence process. **Hypothesis v is therefore strongly supported.**

Comparison with the network-only model (3.1)

The network part of the co-evolution model (effects 3-10) retains the same structural and covariate effects as in 3.1, but alcohol similarity is now expressed via the behavioural dependent variable (`alcohol_dep similarity`).

- **Alcohol similarity (selection):**
`alcohol_dep similarity` = 1.65 (SE = 0.36, $p < 0.001$).
 This is larger than the constant-covariate version (0.50) and shows that the selection effect is robust even after accounting for simultaneous influence. Pupils prefer friends with similar current alcohol consumption.
- **Hypothesis i (indegree popularity):**
 Coefficient = 0.053 ($p < 0.001$), essentially unchanged from 0.056 in the network-only model. Popular pupils remain more attractive as friends.
- **Hypothesis ii (alcohol similarity on tie formation):**
 Now captured by `alcohol_dep similarity` (above). The significance and direction are consistent with the earlier finding.

- **Hypothesis iii (spatial proximity):**
logdist_cov = -0.200 ($p < 0.001$), very close to the previous -0.186. Greater distance still strongly reduces friendship probability.
- **Reciprocity, transitivity, gender homophily:**
All remain strong and significant, with only minor numerical changes.

In summary, the conclusions about hypotheses i–iii are unchanged by incorporating the behavioural dynamics. The joint model confirms that alcohol-based selection and spatial proximity both shape friendship formation, while a simultaneous social influence process drives alcohol convergence among friends.

3.2.5 Selection, influence, or both?

Both effects are positive and statistically significant:

Mechanism	Effect	Estimate	SE	p
Selection	alcohol_dep similarity	1.65	0.36	< 0.001
Influence	average similarity	6.39	1.76	< 0.001

The data therefore provide clear evidence for **both** selection and influence operating simultaneously. Adolescents choose friends based in part on alcohol similarity, and once friendships are established, they influence each other’s drinking habits.

This pattern indicates a *co-evolutionary feedback loop*: alcohol use serves as a basis for friendship sorting, and friendship groups subsequently reinforce similar alcohol consumption through peer influence. Neither a pure selection-only nor a pure influence-only interpretation is sufficient to describe the observed dynamics.

3.3 Improving the model

The GOF diagnostics indicate that the model slightly mis-represents the geodesic distance distribution and the tail of the degree distributions. Two extensions would likely improve the fit:

1. **Geometrically weighted edgewise shared partners (gwespFF).** The current `transTrip` term treats all transitive triplets equally. A `gwespFF` effect with a decay parameter would allow the marginal impact of closing additional triangles to decrease as a dyad already shares many partners, better capturing the observed level of clustering without forcing an over-dense centre. This often improves both the triad census and the geodesic distance fit.
2. **Out-degree activity and popularity.** Adding `outAct` and `inPopSqrt` (or `inPop` with a square-root transformation) can help the model reproduce the variance of the degree distributions, because these terms capture the tendency of some pupils to send (or receive) disproportionately many ties.

3.4 Two additional hypotheses

Hypothesis A: Pupils with the same ethnic background are more likely to become and remain friends.

Operationalisation: If a categorical variable `ethnicity` were available, we would add the `sameX` effect for that covariate. A positive coefficient would indicate homophily.

Pictorial representation: Nodes coloured by ethnicity; ties tend to connect nodes of the same colour.

Hypothesis B: The influence of friends on alcohol consumption is stronger for boys than for girls.

Operationalisation: This could be tested by an interaction between `avSim` and `gender`. In `RSiena` one can include an `effFrom` effect of an actor’s attribute on the rate function? More directly, one can split the data by gender and compare models, or include the `egoX` interaction with `avSim`? A feasible approach

is to define an `egoX * avSim` interaction (available via `includeEffects(..., avSim, interaction1 = "friendship", interaction2 = "alcohol_dep", interaction3 = "gender")`)? RSiena does not directly permit three-way interactions, but one can create an ego-alter alcohol similarity effect that varies by ego-gender using `includeInteraction`. For simplicity, we propose to estimate the model separately for boys and girls and compare the `avSim` coefficients. A significantly larger coefficient for boys would support the hypothesis.

Pictorial representation: Two panels showing a boy and a girl node, with arrows pointing to friends; the boy's arrow is thicker, indicating a stronger influence.

Both hypotheses are theoretically plausible and, given the necessary data, could be tested within the SAOM framework. Once you've run the models, fill in the concrete numbers (coefficients, standard errors, p-values) and adjust the interpretations accordingly. The structure above follows the assignment's point breakdown exactly.